12/4/2025

DERMATOLOGICAL PHARMACOLOGY

TUC-COM
IS-NPMSD
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RECOMMENDED RESOURCES FOR READING

- Katzung: Basic and Clinical Pharmacology
  - Ch 61: Dermatologic Pharmacology
    - bread and butter of topic summary; comprehensive lists of drug classes with derm application

- DiPiro's Pharmacotherapy: a Pathophysiological Approach

focus on new drugs

  - Section 16: Dermatologic Disorders
    - Chapter 113: Acne Vulgaris
    - Chapter 114: Psoriasis
    - Chapter 115: Atopic Dermatitis
    - Chapter 116: Alopecia
    - Chapter e117: Dermatologic Drug Reactions, Contact Dermatitis, and Common Skin Conditions
    - Chapter 156: Melanoma
    - do not read all these chapters unless you want to; recommended as supplemental ref

All these books can be found at: https://libguides.tu.edu/COMbooks/pharmacology

…but you don't have to take my word for it….

LECTURE OUTLINE

Drug lists (on your own)

Your skin is a living thing

- PK review

Drug-induced derm emergencies

Overview of treatments for common conditions

- acne
- psoriasis
- eczema/atopic dermatitis
- neoplasms

Retinoids

Photochemotherapy

Photoprotection: sunscreens

DRUG **CLASSES** TO KNOW

- **Antibiotics**
- cell wall synthesis inhibitors:
  - bacitracin, gramicidin
- protein synthesis inhibitors:
  - neomycin, gentamycin, tetracyclines
  - **clindamycin, erythromycin (DOC in acne)**
  - ozenoxacin, **muciprocin, retapamulin**
    - **effective against MRSA**
- polymyxin B sulfate
- metronidazole (rosacea)
- ivermectin (rosacea, mites)
- sodium sulfacetamide (acne)
- dapsone (acne)

- **Antifungals: topical**
- azoles, e.g.,**miconazole,** clotrimazole, efinaconazole, econazole, ketoconazole, luliconazole, oxiconazole, sertaconazole, sulconazole
- ciclopirox olamine
- naftifine
- terbinafine
- butenafine
- tolnaftate
- nystatin
- amphotericin B

- **Antifungals: oral**
- griseofulvin
- terbinafine
- **fluconazole**
- itraconazole

These drugs are <u>not unique</u> to dermatology: brief coverage only in these slides.

azoles C/I with midazolam and HMG CoA reductase inhibitors

DRUG **CLASSES** TO KNOW, CONT'D

- **Antivirals**
- acyclovir
- valacyclovir
- penciclovir
- famciclovir

- **antiproliferatives and keratinolytic drugs**
- 5-flourouracil
- methotrexate
- salicylic acid
- azaleic acid (note multiple MOA)

- **anti-ectoparasitics**
- insecticides/pediculicide
  - permethrin
  - spinosad
  - ivermectin
    - also antihelminthic
  - lindane
    - note ovicide activity also
  - crotamiton
  - malathion

These drugs are not unique to dermatology: brief coverage only in these slides.

- **immunomodulators** psoriasis, eczema
- TNF-α inhibitors, e.g.,
  - adalimumab
  - etanercept
  - infliximab
- cytokine inhibitors, e.g.,
  - ixekizumab
  - secukinamab
- Toll-like receptor agonists, e.g.,
  - imiquimod (warts, carcinomas)
- calcineurin inhibitors, e.g.,
  - tacrolimus, pimecrolimus
- anti-inflammatories
  - steroid, e.g., corticosteroids
  - non-steroid, e.g., crisaborole
- PDE4 inhibitors
  - apremilast (psoriasis)

DRUGS TO KNOW: DERM-SPECIFIC

- **pigmentation and Vit D modulators**
- hydroquinone, mequinol (hyperpigmentation)
- monobenzone (hyperpigmentation)
  - irreversible depigmentation
- psoralens (vitiligo)
  - trioxsalen
  - methoxsalen
- calcipotriene, calcitriol (psoriasis)

- **retinoids**
- tretinoin
- isotretinoin
- retinoic acid
- adapalene
- tazarotene
- acitrecin

These drugs ARE unique to dermatology so more extensive coverage here.

- **photoblockers** mineral
  - TiO_2, ZnO_2
- **photoabsorbers** chemical
  - homosalate
  - octisalate
  - avobenzone
  - oxybenzone
  - octocrylene
  - octinoxate
  - benzophenones
    - oxybenzophenone
    - dioxybenzone
    - sulizobenzone

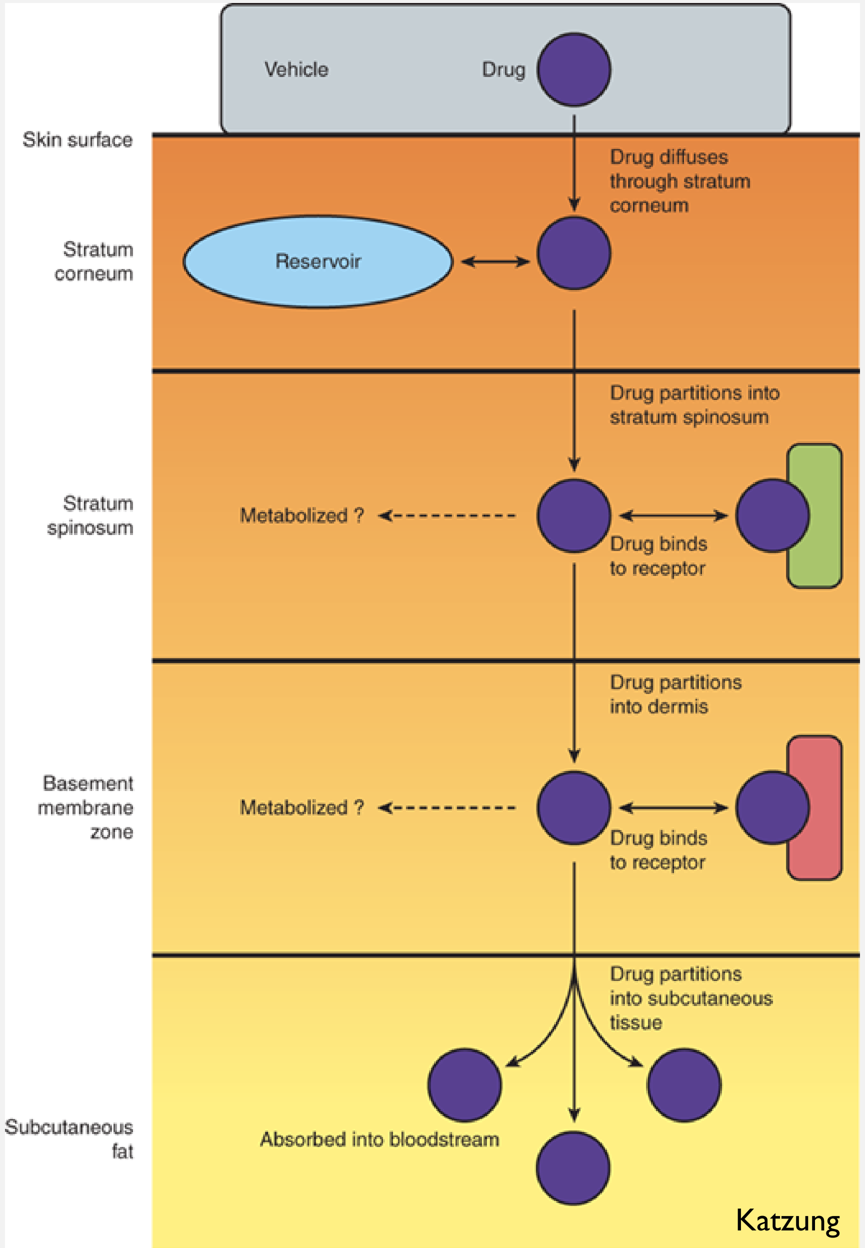
FYI: combo in Neutrogena Beach Defense SPF 70, the sunscreen in my house that I read the ingredient list of while prepping this lecture. I do not endorse or receive anything from Neutrogena ☺

STUDY GUIDE: WHAT TO FOCUS ON

- Know CLASSES of drugs used for common dermatological conditions
  - know some common examples, especially if preferred treatment/DOC
- Explain the most common clinical strategies for treating acne, psoriasis, eczema/atopic dermatitis, and hyperpigmentation
- Recognize DERMATOLOGICAL EMERGENCIES and how to address them
  - SJS, TEN, DRESS, purpura fulminans, AGEP, pyoderma gangrenosum
- Explain RETINOIDS MOA, indications, C/I, and SE
  - e.g., molecular mechanisms of action, which are C/I in pregnancy, retinoid-induced dermatitis and skin irritation
- Explain PSORALEN and PHOTOCHEMOTHERAPY MOA, indications, C/I, and SE
  - remember that psoralens don't work alone; they need to see the light (i.e., are light-activated)
- Explain how SUNSCREENS work
  - Physical/inorganic filtering and chemical/organic filtering agents
  - explain MOA with respect to UV wavelength – distinguish UVA and UVB damage to skin and agents targeting each range
  - explain SPF (it's an adjective, not a noun)

PHARMACOKINETIC CONSIDERATIONS OF THE SKIN

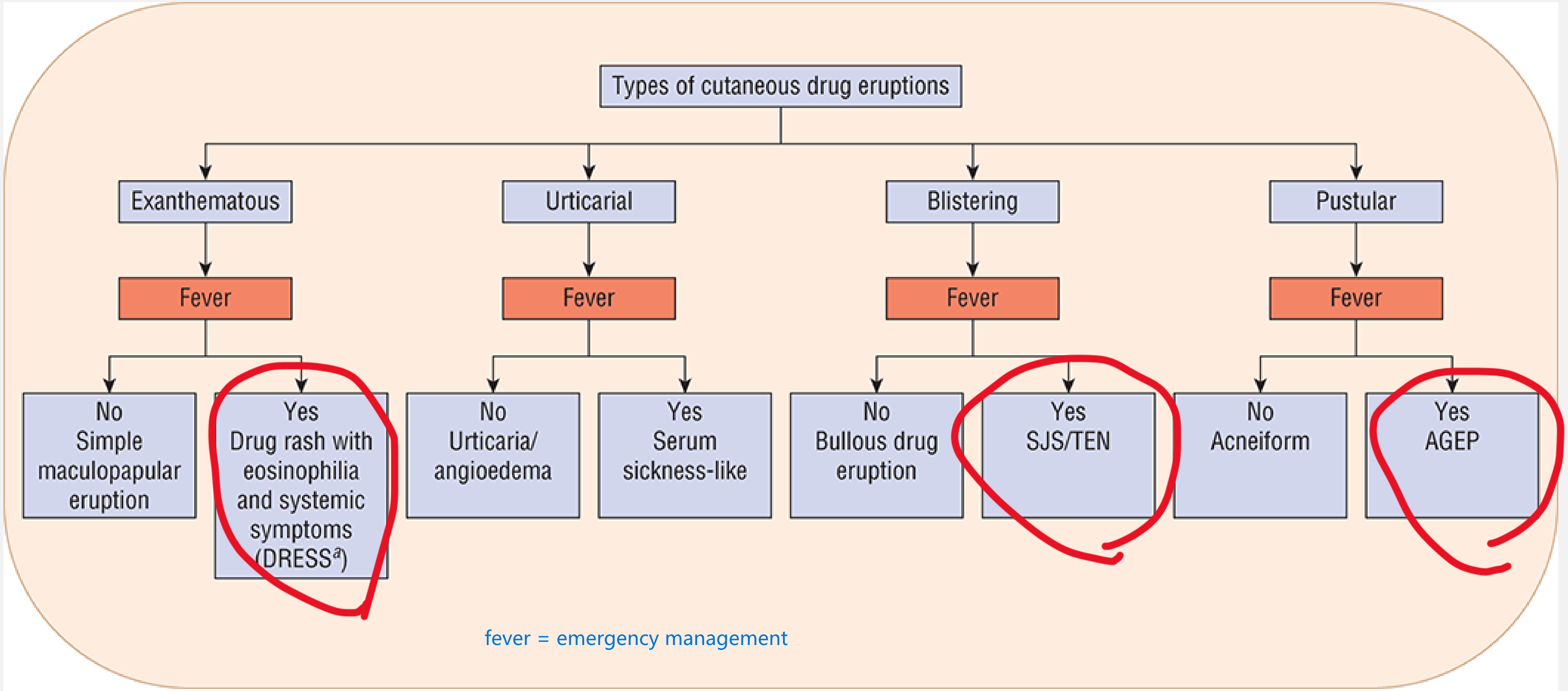
- absorption
  - topical formulations
    - concept note: allows use of agents not indicated for systemic use!
  - thickness of stratum corneum
  - skin hydration
  - occlusion of preparation
- distribution
  - partition coefficient of drug
    - lipophilicity
    - hydrophilicity
- metabolism
  - catalytic activity present in stratum spinosum, basement membrane
- elimination
  - degree of vascularization
    - to carry drug away from site
  - subsequent systemic effects
    - including metabolism and excretion



LOCAL CUTANEOUS REACTIONS TO TOPICAL DRUGS

Reaction Type	Mechanism	Comment
Irritation	Nonallergic	Most common local reaction
Photoirritation	Nonallergic	Phototoxicity; usually requires UVA exposure
Allergic contact dermatitis	Allergic	Type IV delayed hypersensitivity
dermatitis - over days Photoallergic contact dermatitis	Allergic	Type IV delayed hypersensitivity; usually requires UVA exposure
Immunologic contact urticaria	Allergic	IgE-mediated type I immediate hypersensitivity; may result in anaphylaxis
hive min-hrs Nonimmunologic contact urticaria	Nonallergic	Most common contact urticaria; occurs without prior sensitization

DERMATOLOGIC EMERGENCIES: REACTIONS TO MEDICATION



Source: JT DiPiro, GC Yee, LM Posey, ST Haines, TD Nolin, VL Ellingrod. Pharmacotherapy: A Pathophysiologic Approach. 11th Edition.

DERMATOLOGICAL EMERGENCIES TO KNOW

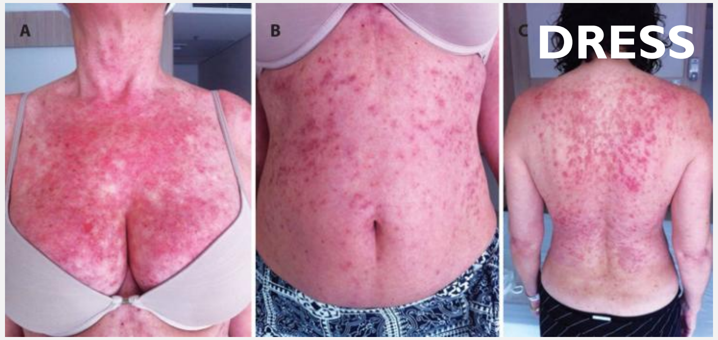
- Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN): allergic response to certain drugs.
- **AN ACUTE LIFE-THREATENING EMERGENCY**
  - common drug "triggers": sulfonamides, penicillins, some anticonvulsants (hydantoins, carbamazepine, barbiturates, lamotrigine), NSAIDs, and allopurinol ethosuximide
  - acute onset (7-14 days of drug exposure)
  - generalized tender or painful bullous formation with accompanying systemic signs and symptoms, including fever, headache, and respiratory symptoms
  - rapid deterioration, including extensive epidermal detachment and sloughing
  - Treatment: **ICU**

https://metro.co.uk/2021/06/02/black-woman-denied-help-as-white-doctors-didnt-recognise-symptoms-14699532/

SJS rxn to **carbamazepine**; drs did not take drug history or recognize SJS

UptoDate

- Drug Rash with Eosinophilia & Systemic Symptoms (DRESS) Syndrome
  - common drugs: allopurinol, sulfonamides, some anticonvulsants (barbiturates, phenytoin, carbamazepine, lamotrigine), and dapsone
  - Not infection; treat with prolonged immunosuppressants



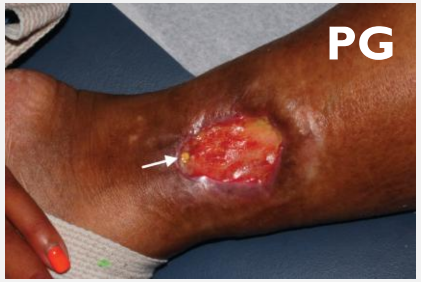
DERMATOLOGICAL EMERGENCIES TO KNOW

- Purpura Fulminans
  - can be caused by meningococcal infection
  - May develop disseminated intravascular coagulation (DIC)
  - treat with broad spectrum antibiotics
- Acute Generalized Exanthematous Pustulosis (AGEP)
  - common drugs: antibiotics, antifungals, , antimalarials, androgenic hormones, some anticonvulsants (e.g., ethosuximide), diltiazem, isoniazid, and lithium.
  - Treat with topical CS or systemic immunosuppressants
- Pyoderma Gangrenosum
  - autoinflammatory ulcerative skin disorder; Not infection or allergic reaction; treat with immunosuppressants (e.g. CS)

no questions about dx from pictures

UptoDate

UptoDate



BOARDS-STYLE EXAMPLE

- A 7-year-old female patient presents to her pediatrician's office, brought by her parent The child's teacher reported that the patient has periods of time where she stares at the wall and seems to “space out." These "space out" episodes last about 10 seconds and occur multiple times a day. During the episodes, she does not lose body tone or have violent movements of the limbs. The patient is not aware of these episodes while they occur and continues with whatever activity she was doing prior to the episode. The patient's medical history is otherwise unremarkable, and she has no known drug allergies. After appropriate testing, she is placed on an appropriate pharmacotherapy treatment regimen. Given the most likely prescribed medication for this condition, which of the following side effects would be the most appropriate for the physician to discuss with the mother?

ethosuximide - anticonvulsar

- A. extrapyramidal symptoms
- B. gingival hyperplasia
- C. pseudomembranous colitis
- D. respiratory depression
- E. Stevens-Johnson syndrome

Bottom Line: Typical absence seizures present with loss of consciousness without loss of body tone, lasting on the average of 10 seconds. Ethosuximide is the preferred treatment. As with other antiepileptics, a potentially serious side effect is Stevens-Johnson syndrome.

COMMON DERMATOLOGICAL CONDITIONS

- acne
  - treatments target oil production, gland occlusion, bacterial growth, inflammation
- psoriasis
  - treatments target autoimmune excess, cell proliferation
- eczema/atopic dermatitis
  - treatments target hypersensitivity
- neoplasms: SCC, BCC, melanoma, cutaneous T-cell lymphoma, others
  - treatments target cancer cell proliferation
- infection/infestation
  - treatments target microbes/parasites
- hyperpigmentation
  - treatment: melanin synthesis inhibitors, melanocyte disruption
- sunburn/photoaging UVB/UVA
  - treatment is supportive; preventive care is photoprotection

TREATMENT OF ACNE

Cutibacterium (formerly *Propionibacterium*) *acnes* proliferation:

- Benzoyl peroxide
- Topical and oral antibiotics
- Azelaic acid

Follicular hyperproliferation and abnormal desquamation:

- Topical retinoids
- Oral isotretinoin
- Azelaic acid
- Salicylic acid

Increased sebum production:

- Oral isotretinoin
- Oral contraceptives
- Spironolactone
- Clascoterone

if hormonal

anti-androgens

Inflammation:

- Oral isotretinoin
- Oral tetracyclines
- Topical retinoids
- Azelaic acid
- Topical dapsone

also for mycobacterium infections like TB

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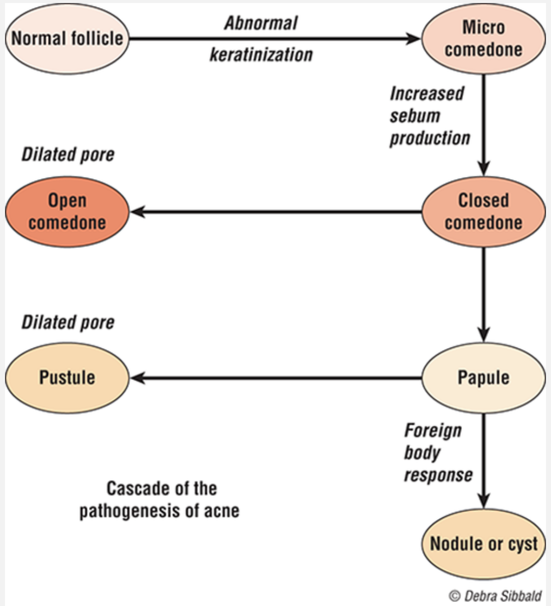

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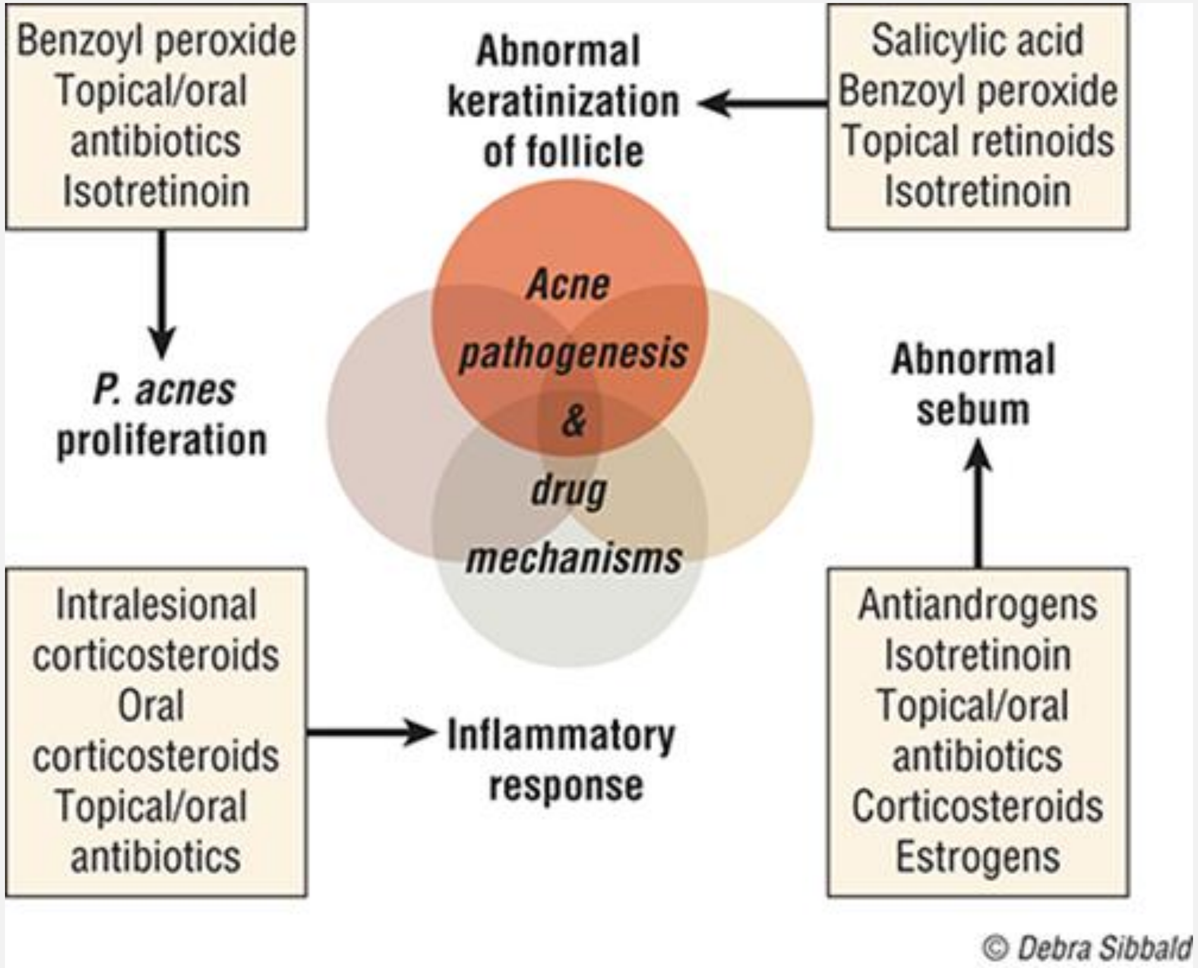

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DiPiro

TREATMENT OF PSORIASIS

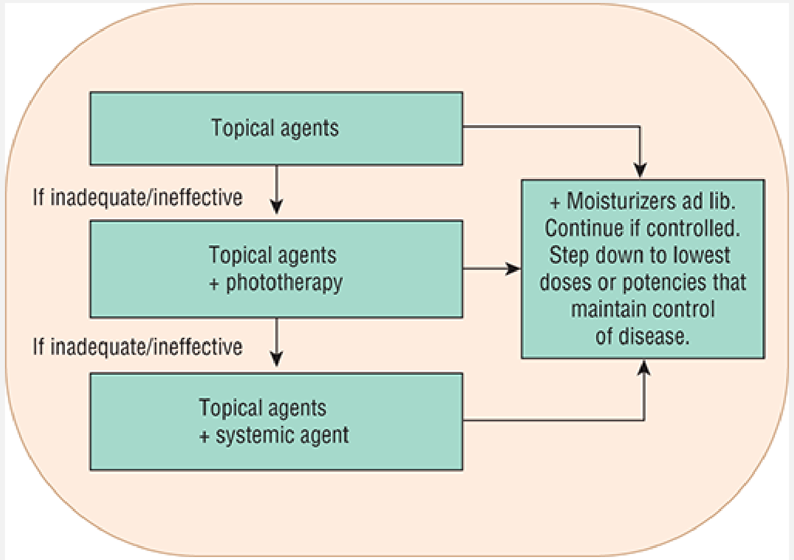
- <u>topical agents</u> (approach of choice for mild-to-moderate)
  - retinoids e.g., tazarotene
  - Vit D analogs calcipotriene, calcitriol
  - corticosteroids e.g., betamethasone
    - reduce after lesions diminish to avoid CS-induced skin atrophy

- <u>phototherapy</u> (moderate-to-severe)
  - UVB narrow or wide "band"

- <u>photochemotherapy</u> (moderate-to-severe)
  - UVA + psoralen

- <u>systemic agents</u> (approach of choice for extensive or moderate-to-severe)
  - methotrexate
  - retinoids, e.g., acitretin
  - cyclosporine
  - biologics
  - note apremilast
    - (PDE4 inhibitor: for mild-to-moderate not controlled by topical agents (UptoDate Jan 2022)

mild



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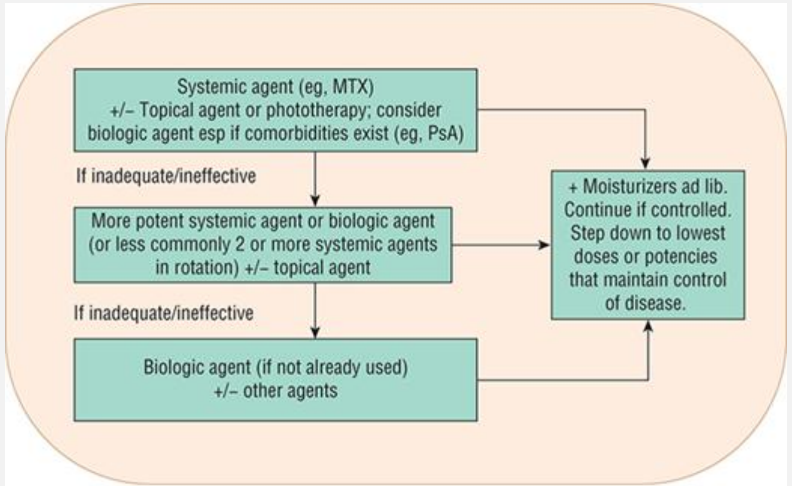

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severe

TREATMENT OF ATOPIC DERMATITIS/ECZEMA

ADULT

(a) Treatment recommendation for atopic eczema: adult

- For every phase, *additional* therapeutic options should be considered
- Add antiseptics / antibiotics in cases of superinfection
- Consider compliance and diagnosis, if therapy has insufficient effect
- Refer to guideline text for restrictions, especially for treatment marked with[1]
- Licensed indication are marked with[2], off-label treatment options are marked with[3]

SEVERE: SCORAD >50 / or persistent eczema		Hospitalization; systemic immunosuppression: cyclosporine A[2], short course of oral glucocorticosteroids[2], dupilumab[1,2], methotrexate[3], azathioprin[3], mycophenolate mofetil[3]; PUVA[1]; alitretinoin[1,3]
MODERATE: SCORAD 25-50 / or recurrent eczema		Proactive therapy with topical tacrolimus[2] or class II or class III topical glucocorticosteroids[3], wet wrap therapy, UV therapy (UVB 311 nm, medium dose UVA1), psychosomatic counseling, climate therapy
MILD: SCORAD <25 / or transient eczema	corticosteroids	Reactive therapy with topical glucocorticosteroids class II[2] or depending on local cofactors: topical calcineurin Inhibitors[2], antiseptics incl. silver[2], silver coated textiles[1]
BASELINE: Basic therapy	lifestyle changes	Educational programmes, emollients, bath oils, avoidance of clinically relevant allergens (encasings, if diagnosed by allergy tests)

PEDIATRIC

(b) Treatment recommendation for atopic eczema: children

- For every phase, *additional* therapeutic options should be considered
- Add antiseptics / antibiotics in cases of superinfection
- Consider compliance and diagnosis, if therapy has insufficient effect
- Refer to guideline text for restrictions, especially for treatment marked with[1]
- Licensed indication are marked with[2], off-label treatment options are marked with[3]

≠	SEVERE: SCORAD >50 / or persistent eczema	Hospitalization, systemic immunosuppression: cyclosporine A[3], methotrexate[3], azathioprin[3], mycophenolate mofetil[1,3]
≠	MODERATE: SCORAD 25-50 / or recurrent eczema	Proactive therapy with topical tacrolimus[2] or class II or III topical glucocorticosteroids[3], wet wrap therapy, UV therapy (UVB 311 nm)[1], psychosomatic counseling, climate therapy
=	MILD: SCORAD <25 / or transient eczema	Reactive therapy with topical glucocorticosteroids class II[2] or depending on local cofactors: topical calcineurin inhibitors[2], antiseptics incl. silver, silver coated textiles
=	BASELINE: Basic therapy	Educational programmes, emollients, bath oils, avoidance of clinically relevant allergens (encasings, if diagnosed by allergy tests)

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note tacrolimus, dupilumab, retinoids, methotrexate

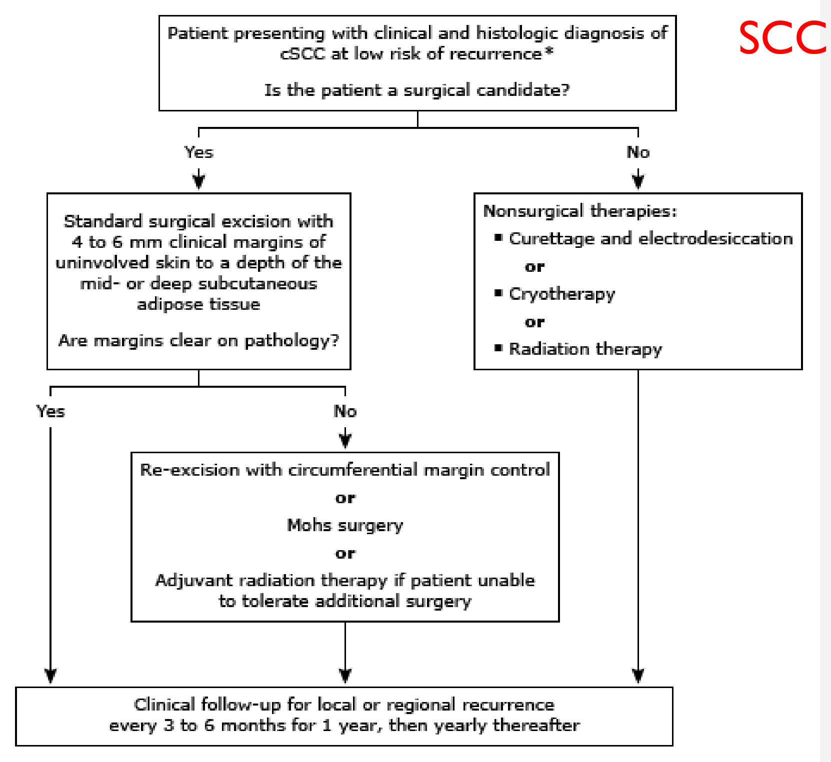
TREATMENT OF DERM NEOPLASMS

- systemic retinoids
  - acitrecin, isotretinoin, alitretinoin, bexarotene
- topical 5-FU

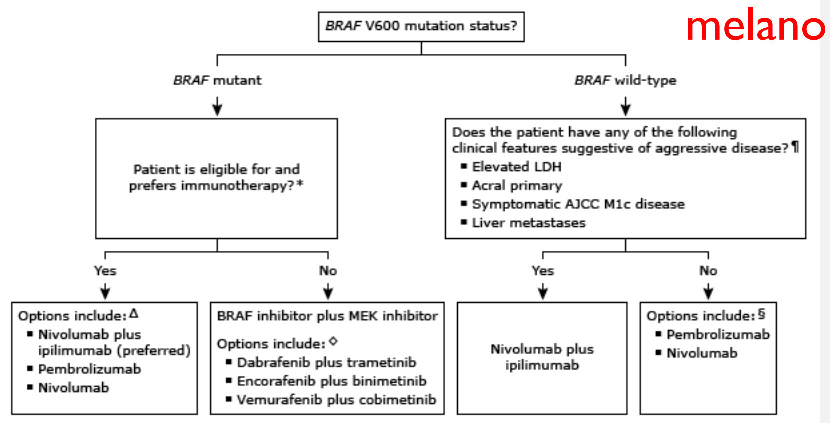
  anti-proliferative agent
  - diffuse SCC, BCC
- Hedgehog pathway inhibitors (HH active in BCC)
  - vismodegib, sonidegib
- imiquimod

  recognize what classes to tx neoplasms
  - Toll-like receptor 7 agonist
  - promotes inflammatory cell infiltration and apoptosis
- other immunomodulators (see flowcharts →)

SCC



melanoma



FYI

UoToDate

HYPERPIGMENTATION CORRECTION

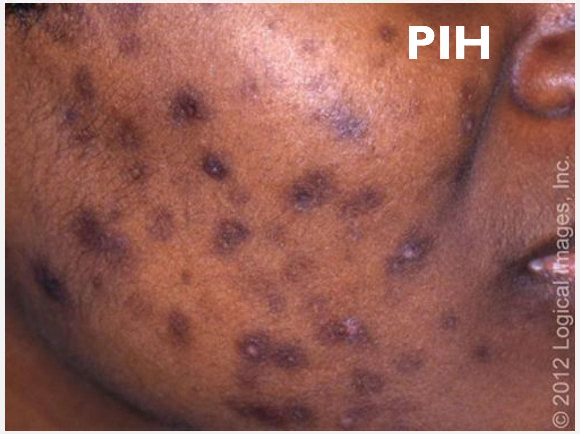
- post-inflammatory hyperpigmentation (PIH)
  - acne, eczematous dermatoses, injury, psoriasis, atopic dermatitis
- melasma
  - photodamage, inflammation, hormone and genetic contributors
- general treatment goal: **inhibition of melanin synthesis/destruction of melanocytes**

- **Treatment approaches:**
  - photoprotection (preventive)
    - broad-spectrum sunscreen preferred
    - containing **iron oxide** (improves UVA-blocking) moves more toward visible spectrum

rx only in us

  - hydroquinone, monobenzone
    - inhibits tyrosinase (melanin synthesis)
    - monobenzone may also destroy melanocytes
  - Vit. C, retinoids
  - azelaic acid (inhibits tyrosinase)
  - thiamidol (inhibits tyrosinase)

- NOT an approved medication: mercury: BANNED WORLDWIDE, NEVER INDICATED, HIGHLY TOXIC!!!!!
  - but may still appear in some cosmetic/unregulated formulations despite being widely banned (WHO)
  - sneaky keywords "antiseptic", "brightening", "beautifying", "purifying", "balancing"
  - see recent public health press releases from MN, TX
  - "People should not buy or use products that may have the following words in ingredient list- "mercury," "mercurio," "calomel," or mercury compounds such as "mercurous chloride.""



UptoDate

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TX Dept of HHS

AE: DEPIGMENTATION AGENTS

- hydroquinone:
  - irritation, abnormal repigmentation (esp in sun-exposed skin), exogenous ochronosis, potential carcinogen (but not established in humans) only in animals
- other skin lightening formulations:
  - corticosteroids: skin atrophy, striae, HPA suppression
  - mercury, arsenic: highly toxic

QUICK REVIEW

- What aspect(s) of acne occurrence do retinoids help to treat?

  all 4

- What drug class would you most likely use to treat:
  - candidiasis?
  - ringworm?
  - head lice?
  - herpes sore?

- A patient presents to the ER with a high fever and extensive blistering on the face, neck, chest, and arms, about 7 days after starting a course of SMX-TMP to treat a lingering URTI. What condition do you suspect and is it serious?

QUICK REVIEW ANSWERS

- What aspect(s) of acne occurrence do retinoids help to treat?
  - bacterial proliferation, inflammation, abnormal keratinization/desquamation

- What drug class would you most likely use to treat:
  - candidiasis?
    - antifungal
  - ringworm?
    - antifungal
  - head lice?
    - pediculicide
  - herpes sore?
    - antiviral

tx dependent on derm condition

- A patient presents to the ER with a high fever and extensive blistering on the face, neck, chest, and arms, about 7 days after starting a course of SMX-TMP to treat a lingering URTI. What condition do you suspect and is it serious?
  - note fever, plus symptoms consistent with Stevens-Johnson syndrome (SJS) or toxic epidermal necrolysis (TEN); YES IT IS VERY SERIOUS

LECTURE OUTLINE

~~Drug lists (on your own)~~

~~Your skin is a living thing~~

- ~~PK review~~

~~Drug-induced derm emergencies~~

~~Overview of treatments for common conditions~~

- ~~acne~~
- ~~psoriasis~~
- ~~eczema/atopic dermatitis~~
- ~~neoplasms~~

Retinoids

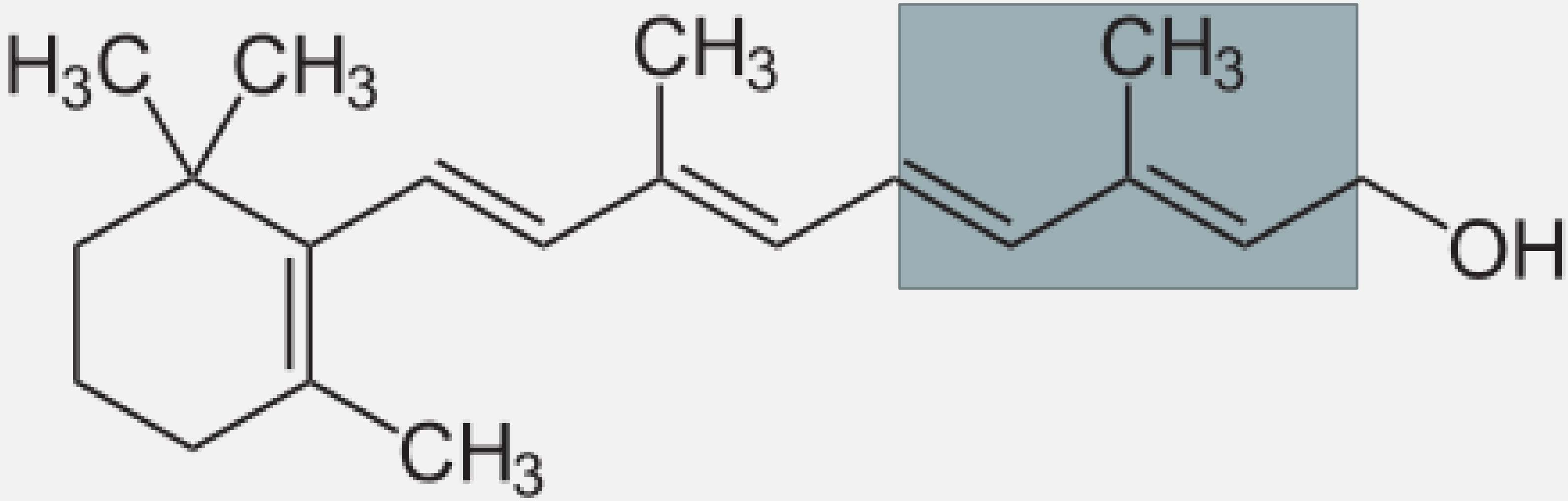
Photochemotherapy

Photoprotection: sunscreens

RETINOID STRUCTURE

will not ask about structure

Isoprene Unit

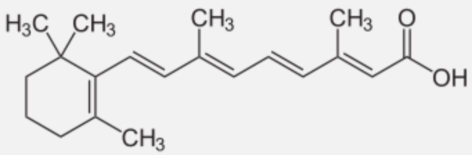
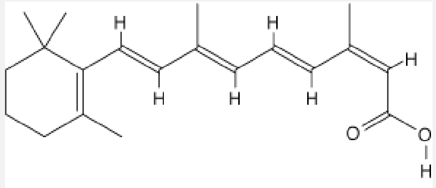
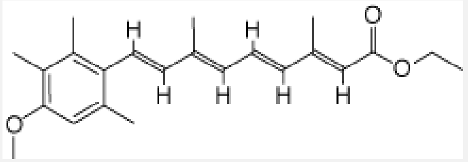
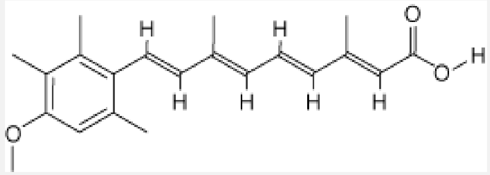
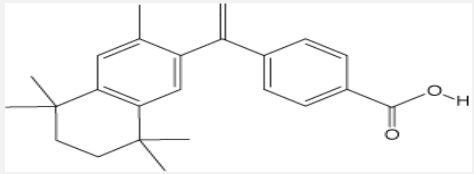


Beta-ionone ring **Isoprenoid: Multiple Isoprene Units**

Vitamin A: retinol

TYPES OF RETINOIDS

- **1st generation retinoids** include **retinol** (vitamin A), **tretinoin** (all-*trans*-retinoic acid; vitamin A acid), **ISOTRETINOIN** (13-*cis*-retinoic acid), and **ALITRETINOIN** (9-*cis*-retinoic acid).
- **2nd generation retinoids** (aka *aromatic retinoids*) include **acitretin** and **etretinate**.
  - note very long half lives
- **3rd generation retinoids** include **tazarotene**, **bexarotene**, and **adapalene**.
  - note optimized for receptor-selective binding

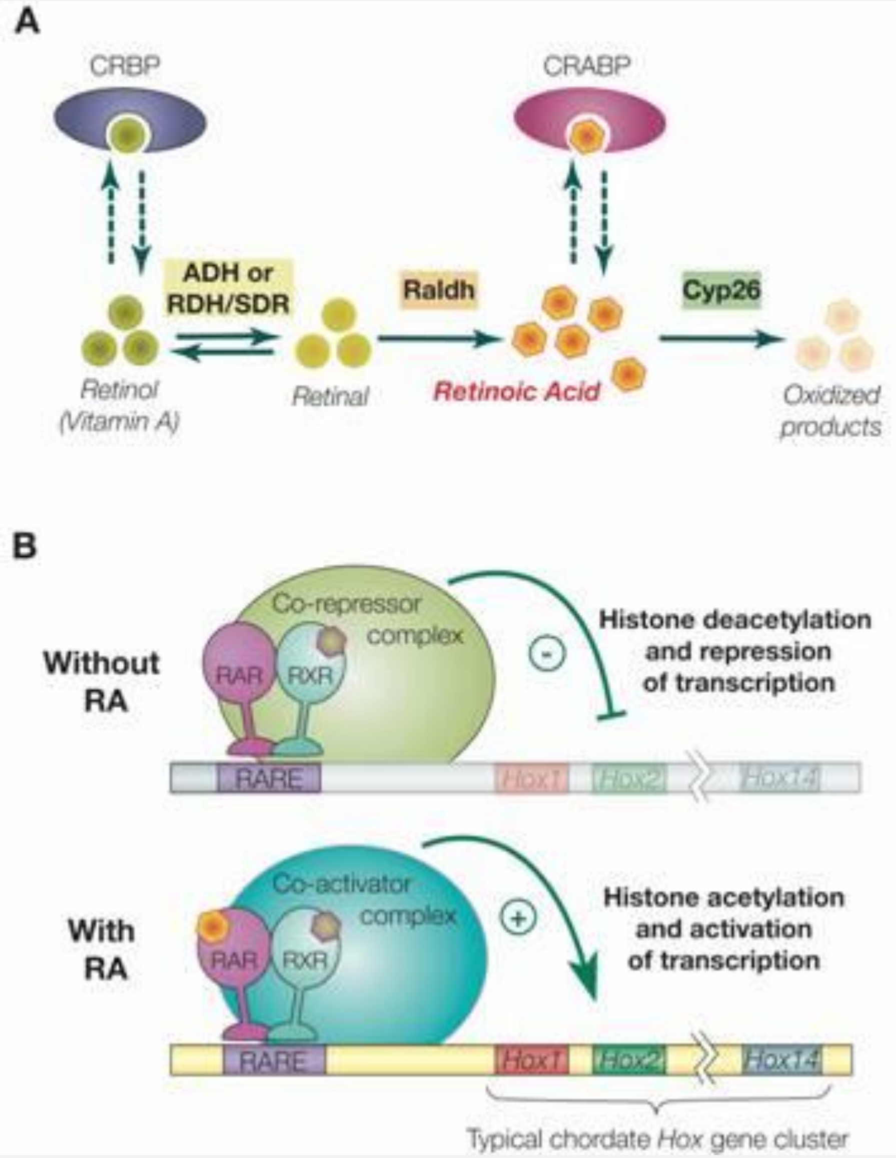
retinoids	STRUCTURE	$t_{1/2}$
retinoic acid		1-2 hr
Isotretinoin		10-20 hr
Etretinate		120 days
Acitretin		50 hours
Bexarotene		7-9 hours

note very long $t_{1/2}$

RETINOIDS MOA: TRANSCRIPTIONAL REGULATION

- **Receptor Activation**
  - **RAR- Retinoic Acid Receptor**
    - Binding by retinoids affects cell proliferation signals*
  - **RXR- Retinoid X Receptor**
    - Binding by retinoids activates apoptosis*
  - **VDR- Vitamin D Receptor**
    - like RAR, VDR forms a heterodimer with RXR; complex tnx outcomes
- **RAR and RXR both have α, β, γ forms**
- RAR
  - drives cell differentiation/proliferation*
    - **Tretinoin, isotretinoin, adapalene, tazarotene**
- RXR
  - drives apoptosis*
    - **Bexarotene, alitretinoin**

* IN GENERAL, not in all cases



Int J Biol Sci 2006; 2(2):38-47. doi:10.7150/ijbs.2.38

RETINOIDS: CLINICAL APPLICATIONS

- **Photoaging**
  - UV irradiation of human skin damages the dermal matrix in a complex manner that is not completely understood.
  - However, topical **tretinoin** partially reverses **collagen deficiency, improves vascularization, and decreases further matrix degradation** induced by solar UV

- **Acne**
  - Retinoids can improve **acne scars** and reduce **postinflammatory hyperpigmentation (PIH)**.

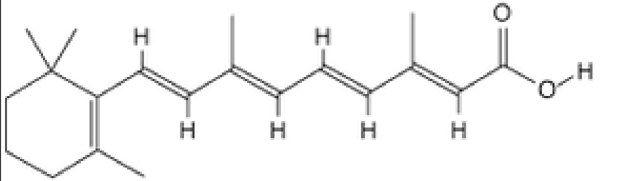
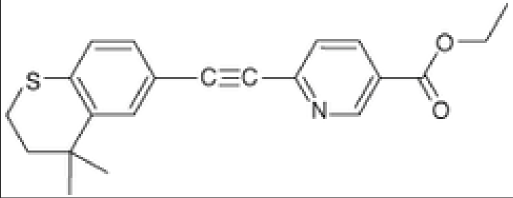
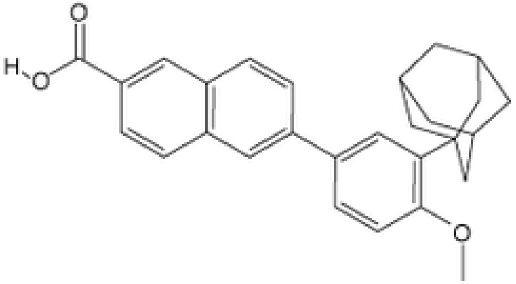
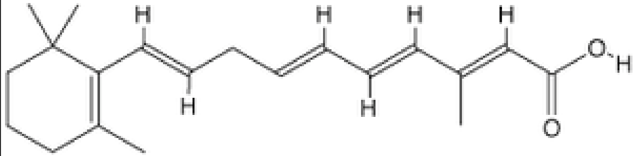
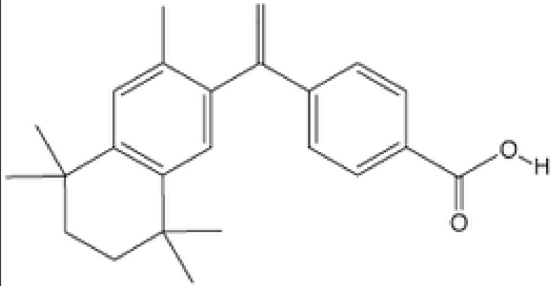
RETINOIDS: CLINICAL APPLICATIONS CONT'D

- **Psoriasis**
  - Retinoids are known to modulate **epidermal proliferation** and **differentiation** and to have **immunomodulatory** and **anti-inflammatory activity**.
  - **Soriatane** (Acitretin) is the only oral retinoid approved by the FDA specifically for treating **psoriasis**. The exact way it works to control psoriasis is unknown.

- **Dermatological Cancers**
  - The oral retinoid **isotretinoin** (Roaccutane) improves wrinkles and other sun-induced skin damage, while **actinic keratoses** and **basal cell carcinomas** have been treated with the topical retinoid **tretinoin**.

TOPICAL RETINOIDS

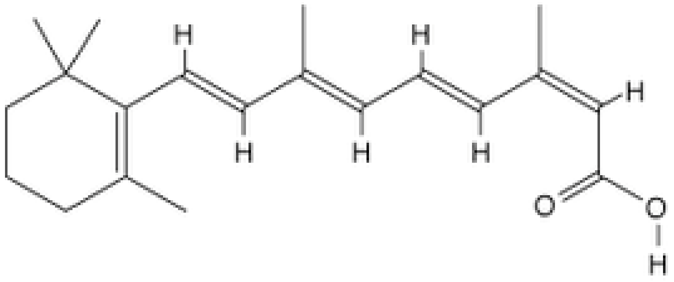
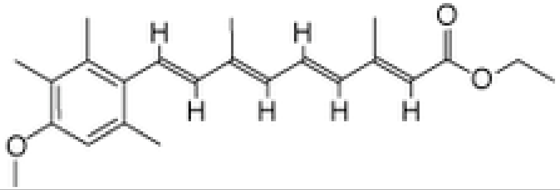
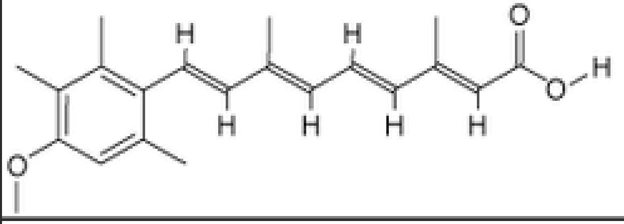
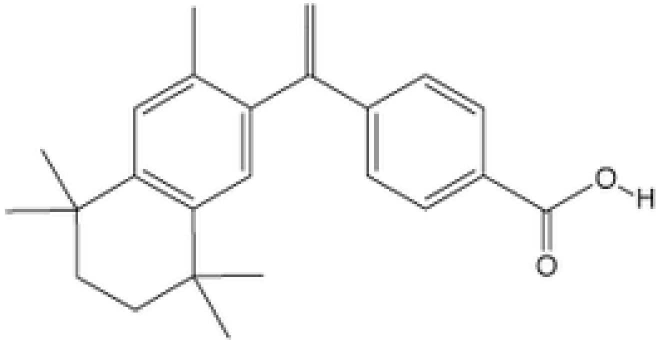
C = no evidence, D = some know effect, X = C/I in pregnancy

DRUG	FORMULATION	RECEPTOR SPECIFICITY	PREGNANCY CATEGORY	STRUCTURE	IRRITANCY
Tretinoin	0.02%, 0.025%, 0.05%, 0.1% cream .01%, 0.025%, 0.04% gel 0 0.05%, 0.1% solution	RAR-α, RAR-β, RAR-γ	C		++
Tazarotene	0.05%, 0.1% cream 0.05%, 0.01% gel	RAR-α, RAR-β, RAR-γ	X		++++
Adapalene	0.1%, 0.3% cream 0.1%, 0.3% gel 0.1% solution	RAR-β, RAR-γ	C		+
Alitretinoin	0.1% gel	RAR-α, RAR-β, RAR-γ	D		++
Bexarotene	1% gel	RXR-α, RXR-β, RXR-γ	X		+++

note: irritancy affects pt compliance

note: **not** all topical retinoids category X

SYSTEMIC RETINOIDS

DRUG	STRUCTURE	RECEPTOR SPECIFICITY	STANDARD DOSING RANGE	$t_{1/2}$
Isotretinoin		No clear receptor affinity	0.5-2 mg/kg/day	10-20 hours
Etretinate		RAR-α, RAR-β, RAR-γ	0.25-1 mg/kg/day	80-160 days
Acitretin		RAR-α, RAR-β, RAR-γ	0.5-1 mg/kg/day	50 hours
Bexarotene		RXR-α, RXR-β, RXR-γ	300 $mg/m^2/day$	7-9 hours

note very long half lives

RETINOIDS ADVERSE EFFECTS

- **Topical**
  - UV sensitivity; sunburn
  - Desquamation (peeling)
  - Irritation: dry-feeling skin, burning/stinging

- **Systemic**
  - **Retinoid toxicity**
    - Dry skin, nosebleeds
    - Hair loss
    - IBS trigger
    - Muscle and joint pain (myalgia, arthralgia)
  - **Potent teratogen:** pregnancy category X
    - RAR/RXR control many aspects of embryonic development
  - **Hypothyroidism**
    - RXR-dependent *TSH* gene suppression

RETINOIDS ADVERSE EFFECTS

- Oral (systemic) retinoids are known teratogens: **absolute C/I during pregnancy**
  - effective contraception MUST be used concurrently with systemic retinoids.

- Ok, but what about topical retinoids?
  - some evidence that topical retinoids may also be teratogenic
    - E.g., topical tretinoin (Stieva-A) is "not recommended during pregnancy or in women of childbearing years without the proper use of an effective method of contraception." [mfr label]
    - Nonclinical reproductive toxicity studies have detected developmental toxicity at doses ≥80-fold of the anticipated topical tretinoin clinical dose. Although systemic absorption from topical tretinoin is between 1% to 6% when properly used, risk cannot be excluded

note: **not** all topical retinoids category X

note: **all systemic** retinoids are pregnancy category X

DiPiro

BOARDS-STYLE EXAMPLE

A 13-year-old female patient presents to her primary care physician for a follow-up of her treatment for acne vulgaris. The patient is currently using benzoyl peroxide, topical retinoids, and erythromycin. Unfortunately, she does not seem to be responding to the current treatment. The patient is placed on treatment with oral retinoids. Prior to initiating therapy, she and her parents are warned of the common side effects of this treatment, which include:

A. dry skin and muscle pains
B. encephalitis
C. increased incidence of cholangiocarcinoma
D. increased incidence of hepatocellular carcinoma
E. pancreatitis

Bottom Line: Oral isotretinoin produces predictable and manageable side effects that are, for the most part, reversible on discontinuation of therapy. Most are similar to those seen in high dose vitamin A therapy that includes dry cracked lips, xerosis of the skin, mucous membranes and eyes, myalgia and arthralgia, which tend to be transient and dose-related to exercise, and elevated levels of lipids and liver enzymes have been associated with therapy.

QUICK REVIEW

- How do 3rd-gen retinoids tend to differ from previous generations? Give one example of a 3rd-gen retinoid.

- What is the two-word description of retinoids MOA?

- A primarily RAR-interacting retinoid probably drives which of the following, apoptosis or differentiation?

- What is an absolute contraindication for oral etretrinate?

  pregnancy

QUICK REVIEW

- How do 3rd-gen retinoids tend to differ from previous generations? Give one example of a 3rd-gen retinoid.
  - greater receptor selectivity; e.g., tazarotene, bexarotene, adapalene
- What is the two-word description of retinoids MOA?
  - transcriptional regulation
- A primarily RAR-interacting retinoid probably drives which of the following, apoptosis or differentiation?
  - differentiation
- What is an absolute contraindication for oral etretrinate?
  - pregnancy or plans to become pregnant

LECTURE OUTLINE

~~Drug lists (on your own)~~

~~Your skin is a living thing~~

~~PK review~~

~~Drug-induced derm emergencies~~

~~Overview of treatments for common conditions~~

~~acne~~

~~psoriasis~~

~~eczema/atopic dermatitis~~

~~neoplasms~~

~~Retinoids~~

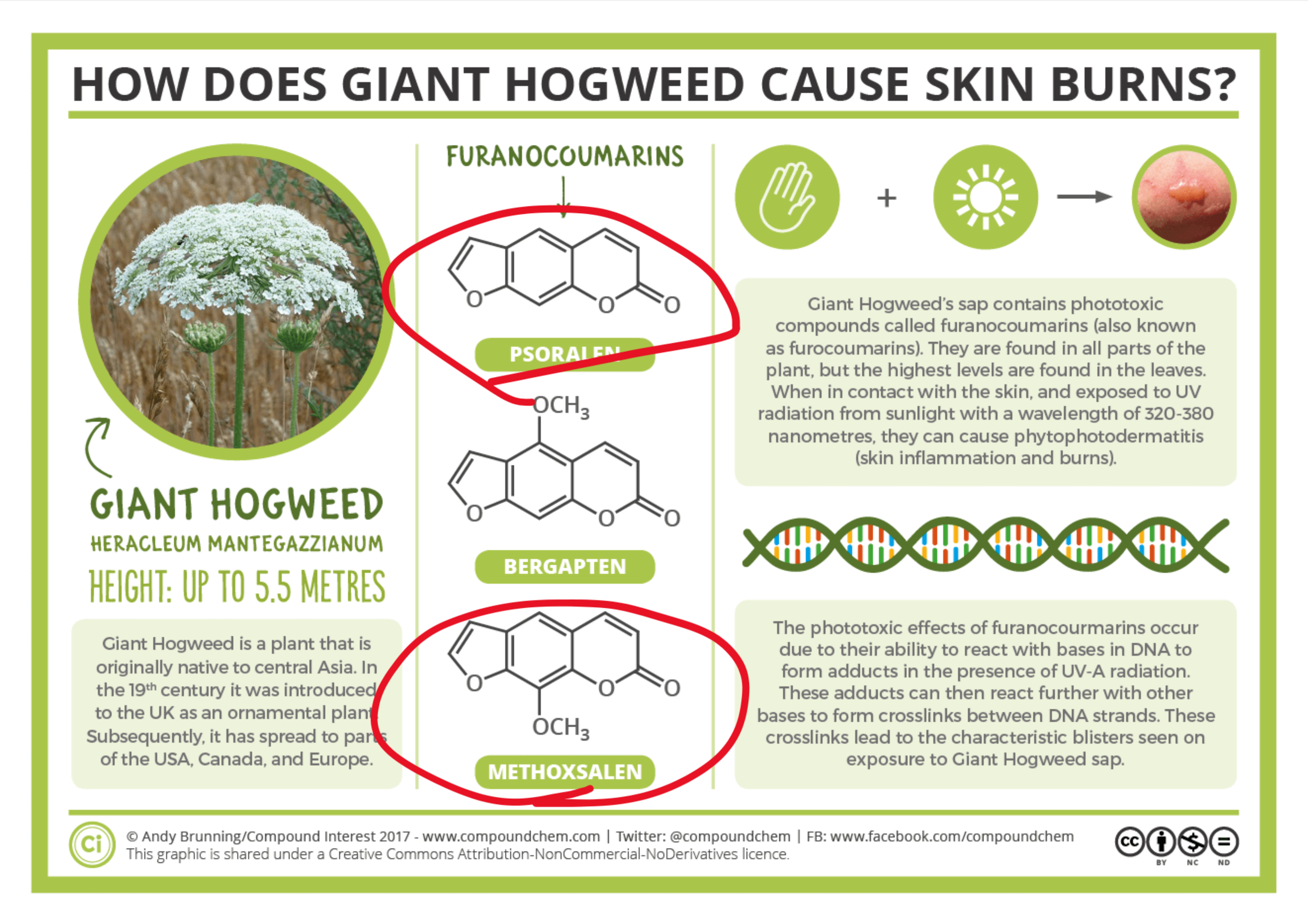
Photochemotherapy

Photoprotection: sunscreens

FYI: WHEN NATURE IS NOT YOUR FRIEND

related furanocoumarin-containing plants:

- carrot
- parsnip
- Queen Anne's lace
- celery
- dill
- fennel
- parsley

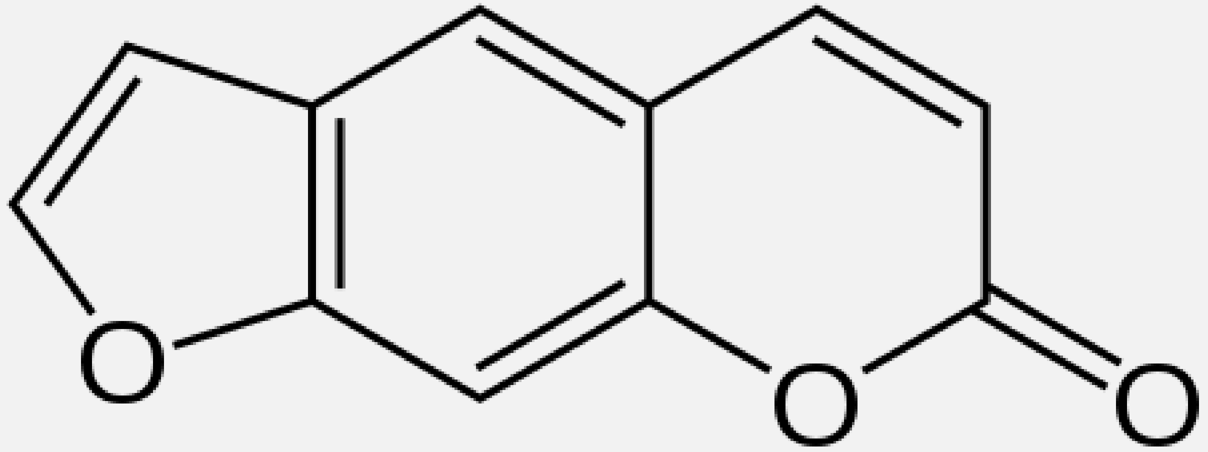


interact with light to cause cell damage

PSORALEN COMPOUNDS

- **Psoralen** (also called psoralene) is the parent compound in a family of natural products known as furanocoumarins.

main agent of photochemtherapy



- It is widely used in P-UVA (psoralen + UVA) treatment for e.g., psoriasis, eczema, vitiligo, and cutaneous T-cell lymphoma.

PHOTOCHEMOTHERAPY = UV + PSORALEN

study q: which type of UV is this?

- UV Light (320-340nM) **in conjunction** with **photosensitizing** agents (**psoralens**)
  - Oral delivery followed by UV exposure

wavelengths and types of rxns

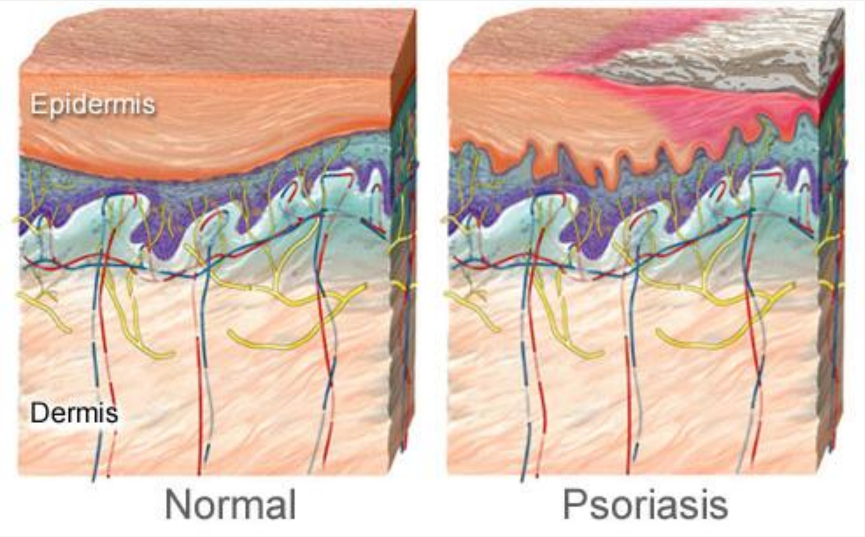
study q: which type rxn is this?

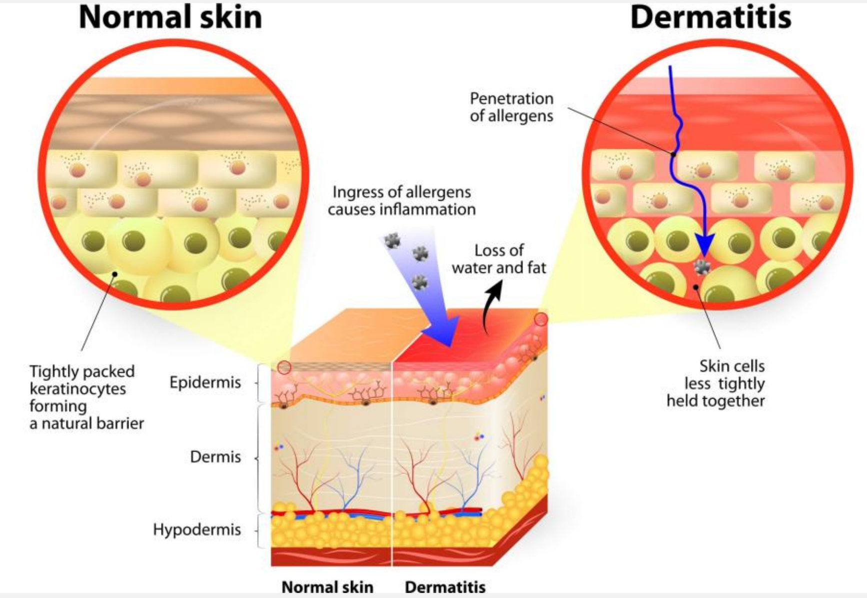
- MOA: UV-activated drugs **intercalate** into skin cell DNA, promoting cell alteration and/or **apoptosis (cell death).**
  - Also induces ROS production, which is pro-apoptotic.

study q: which type rxn is this?

- **Indications**:
  - **Psoriasis**
  - **Atopic Dermatitis**
  - **Skin Cancers**
  - **hypopigmentation correction, vitiligo**
    - vitiligo: autoimmune destruction of melanocytes
    - Non-pigmented patches of otherwise normal skin

Winnie Harlow, Cosmopolitan



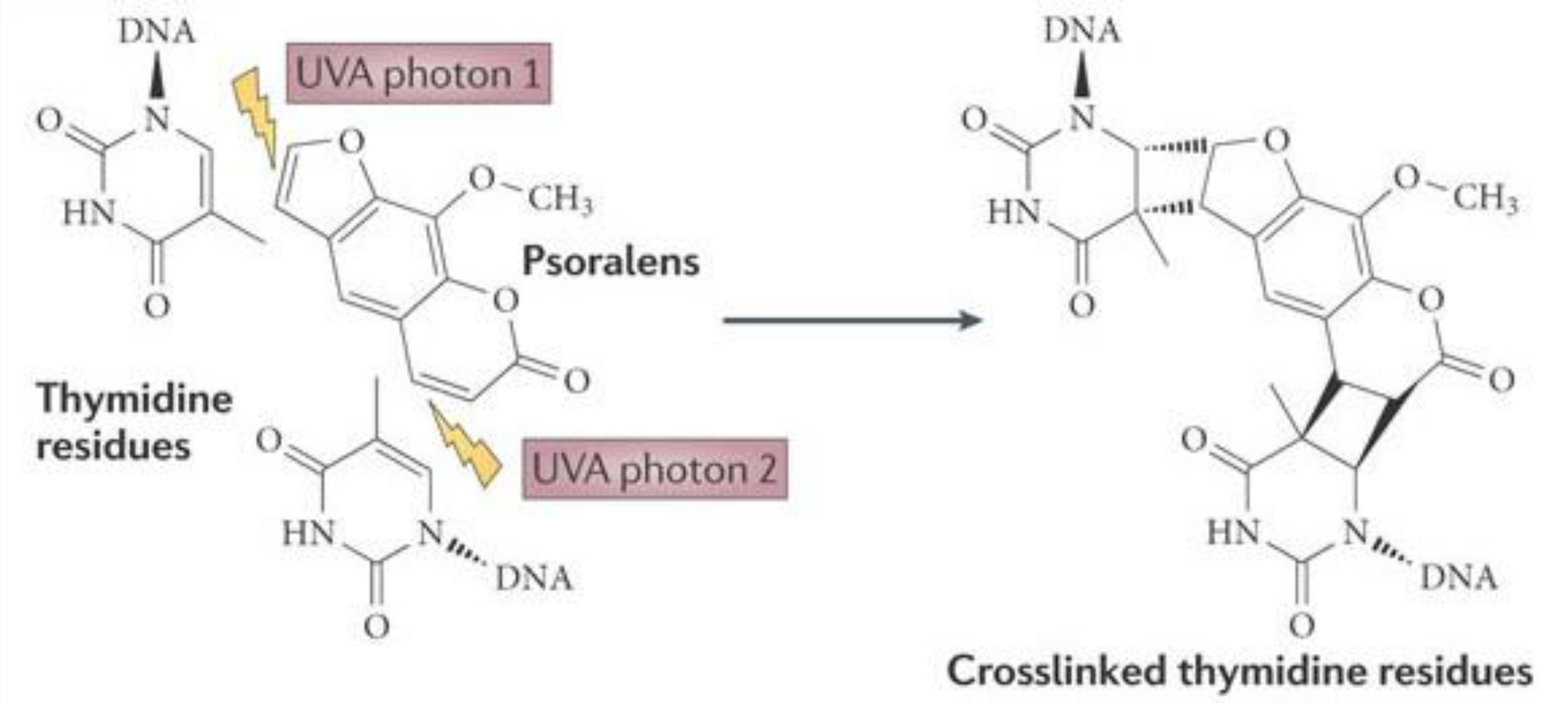


PSORALENS MOA TYPE I : DNA INTERCALATION

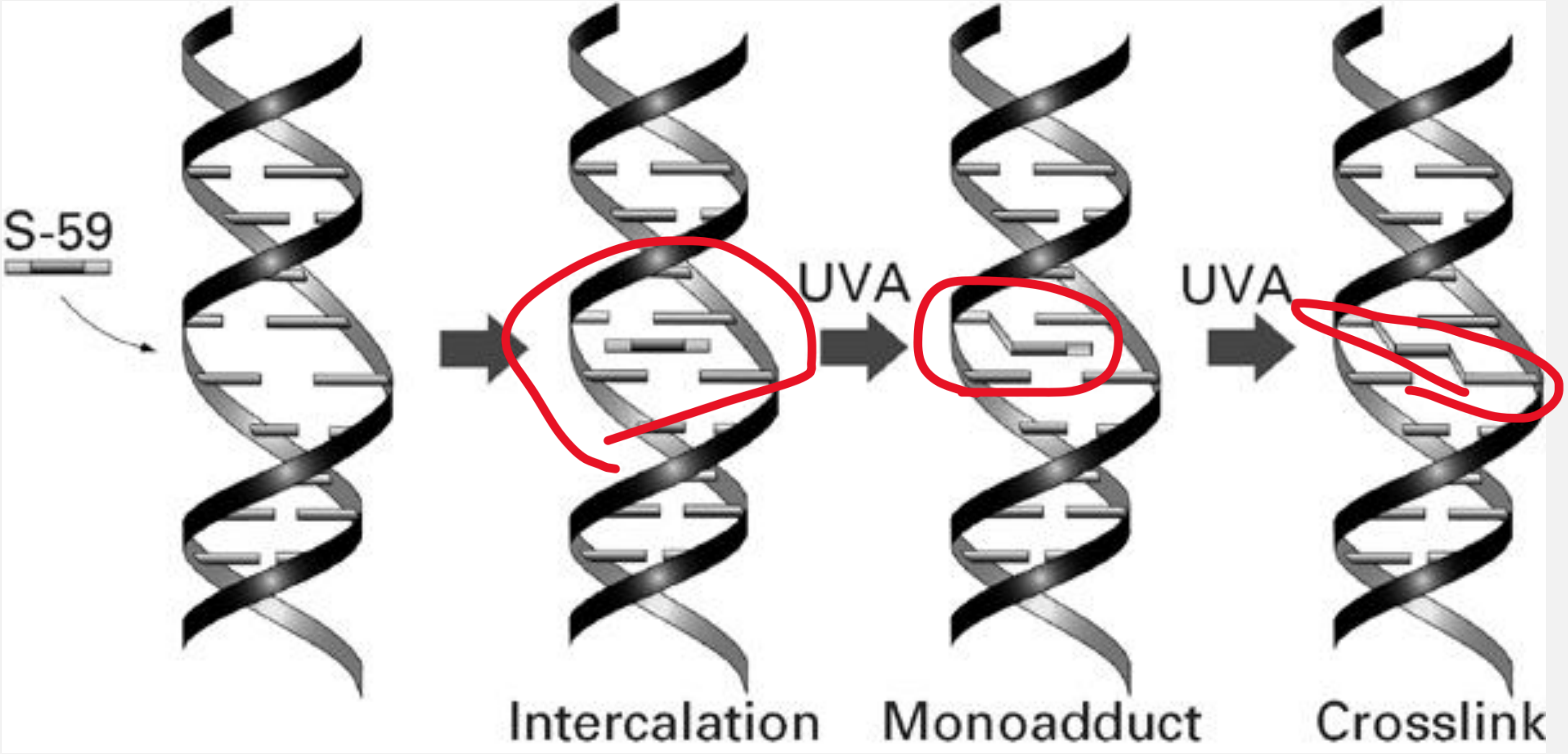
disrupt cell division

- UV photo-activation reaction with **psoralens**.

more than you need to know study q: which psoralen is shown here? hint- see FYI slide

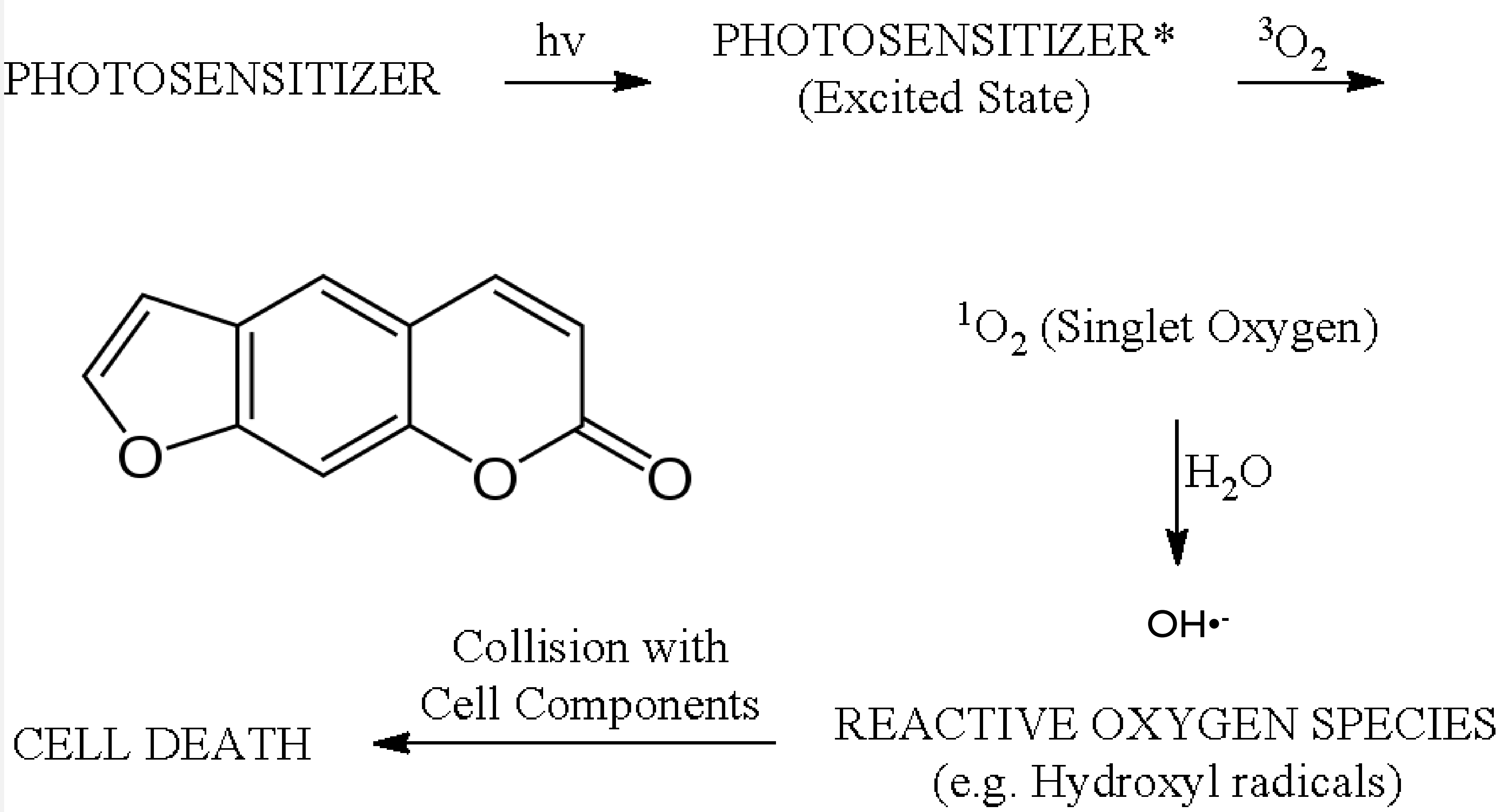


PSORALENS TYPE I REACTION: ANOTHER VIEW



Bone Marrow Transplantation (2004) 33, 1–7

PSORALENS MOA TYPE II : ROS PRODUCTION



AE: PSORALEN AND DERIVATIVES

- psoralens: amplified photosensitivity in combination with other photosensitizing drugs
  - e.g., phenothiazines, thiazides, sulfonamides, NSAIDS, sulfonylureas, tetracyclines, and benzodiazepines
  - Phototoxicity
    - Dry skin, abnormal pigmentation, pruritis, HSV activation
    - Edema, blistering

LECTURE OUTLINE

~~Drug lists (on your own)~~

~~Your skin is a living thing~~

~~PK review~~

~~Drug-induced derm emergencies~~

~~Overview of treatments for common co~~

~~acne~~

~~psoriasis~~

~~eczema/atopic dermatitis~~

~~neoplasms~~

~~Retinoids~~

~~Photochemotherapy~~

Photoprotection: sunscreens

IF YOUR EYES SAW UV, YOU WOULD SEE THAT SUNSCREEN WORKS

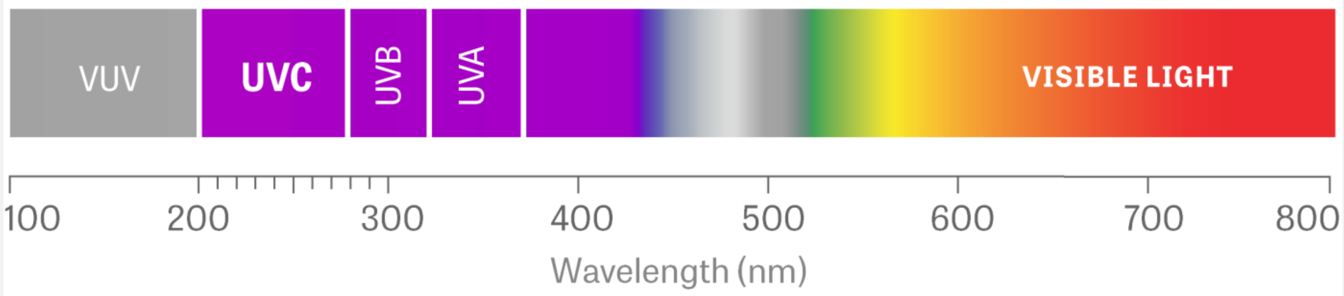
study q: what does this tell you about the mechanism of action of sunscreens?

UV rays not getting reflected back

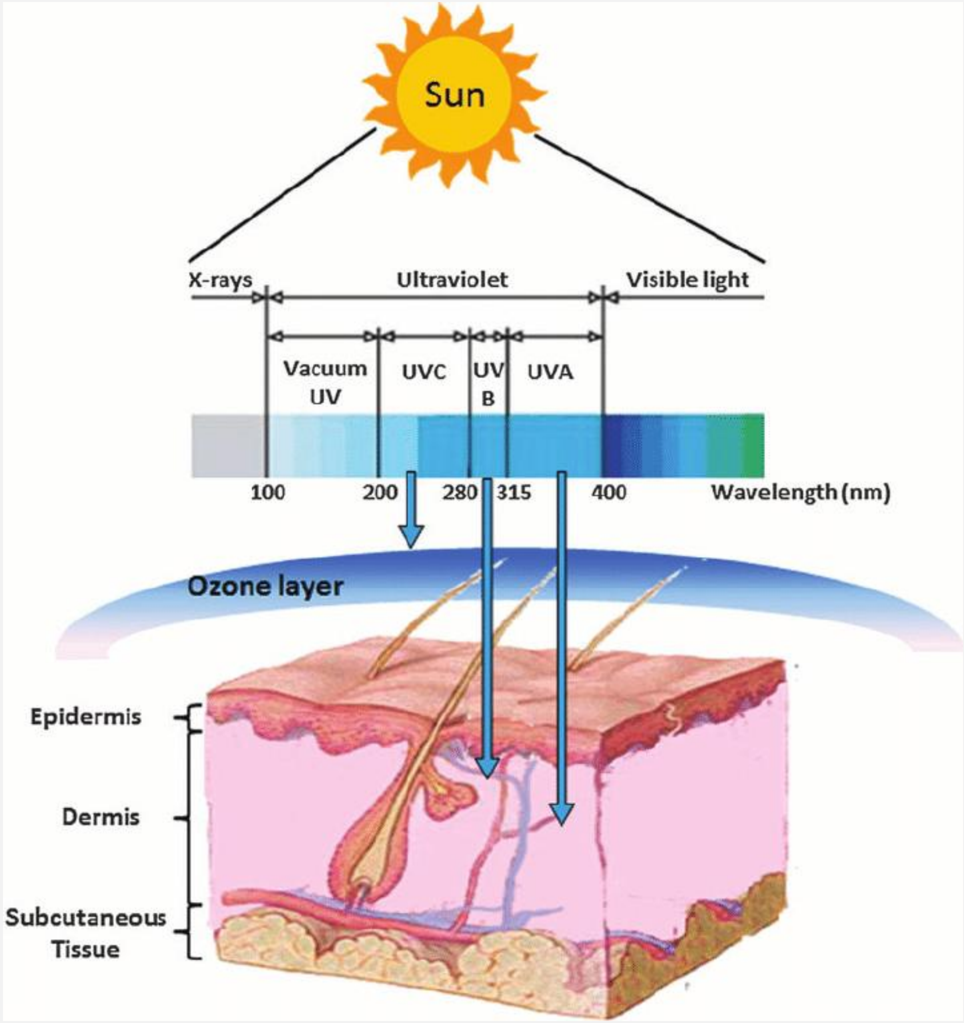
regular camera: detects visible light reflected from surfaces

UV camera: detects UV reflected from surfaces

THE UV SPECTRUM: NOT VISIBLE TO HUMANS



- UV-A: 315-400 nm (A = Aging)
  - longest UV nm, deepest skin penetration- contributes to photoaging, cancer
  - most UVA reaches Earth's surface
- UV-B: 280-315 nm (B = Burning)
  - largest contributor to sunburn, also contributes to cancer
  - despite ~90% absorbed in atmo
- UV-C: ~200-280 nm (C = Captured)
  - ~100% solar UVC absorbed (captured) in atmo O_3 layer, BUT
  - used in laboratory and sterilizing applications
  - also produced by arc welders, Hg lamps

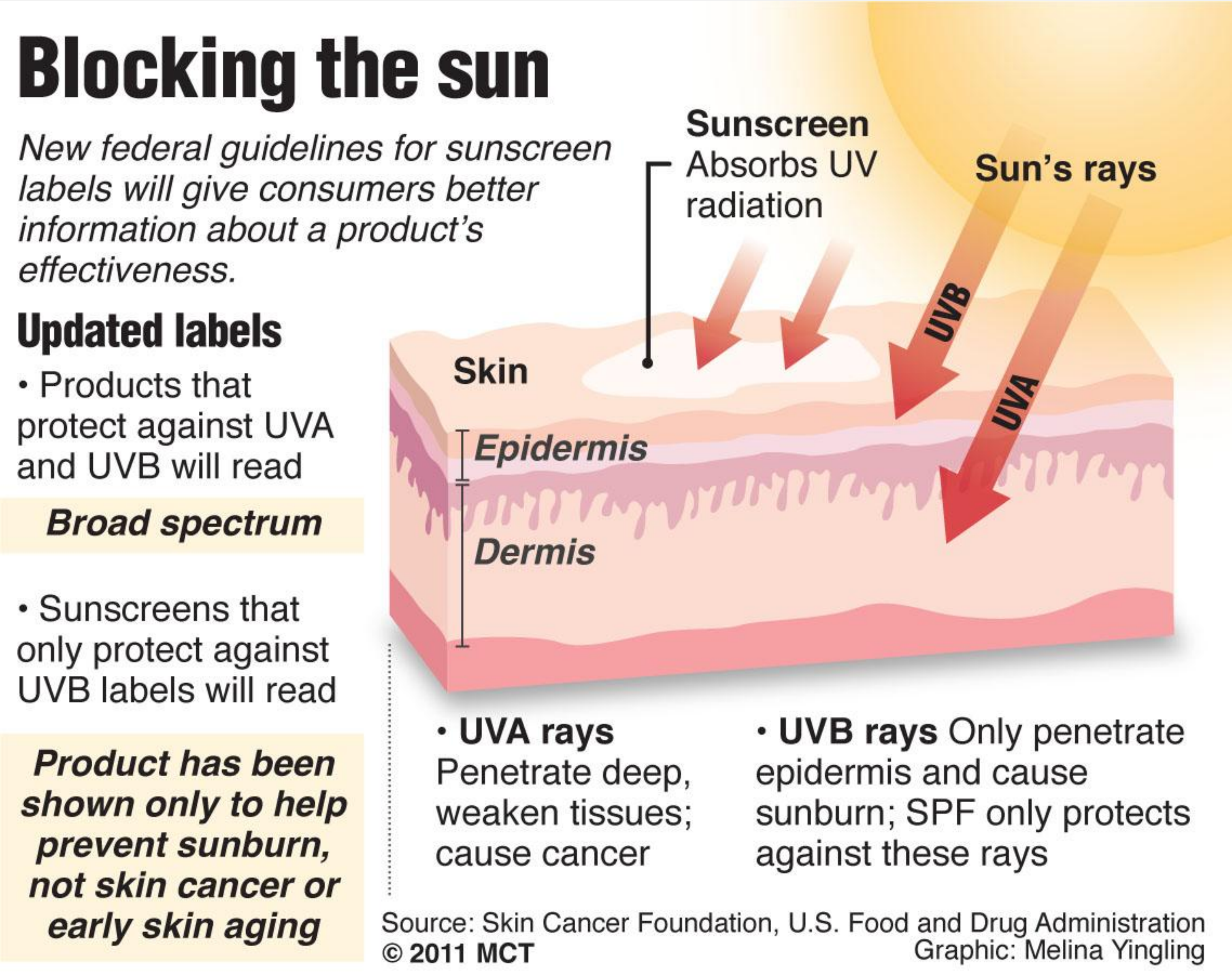


DOI:10.1089/wound.2012.0366

don't need to memorize numbers but know shortest, longest, etc.

MOST CONSUMERS DO NOT UNDERSTAND PHOTOPROTECTION

why? until recently, few products distinguished between UVA and UVB blocking ability; better labeling is now required by the FDA:



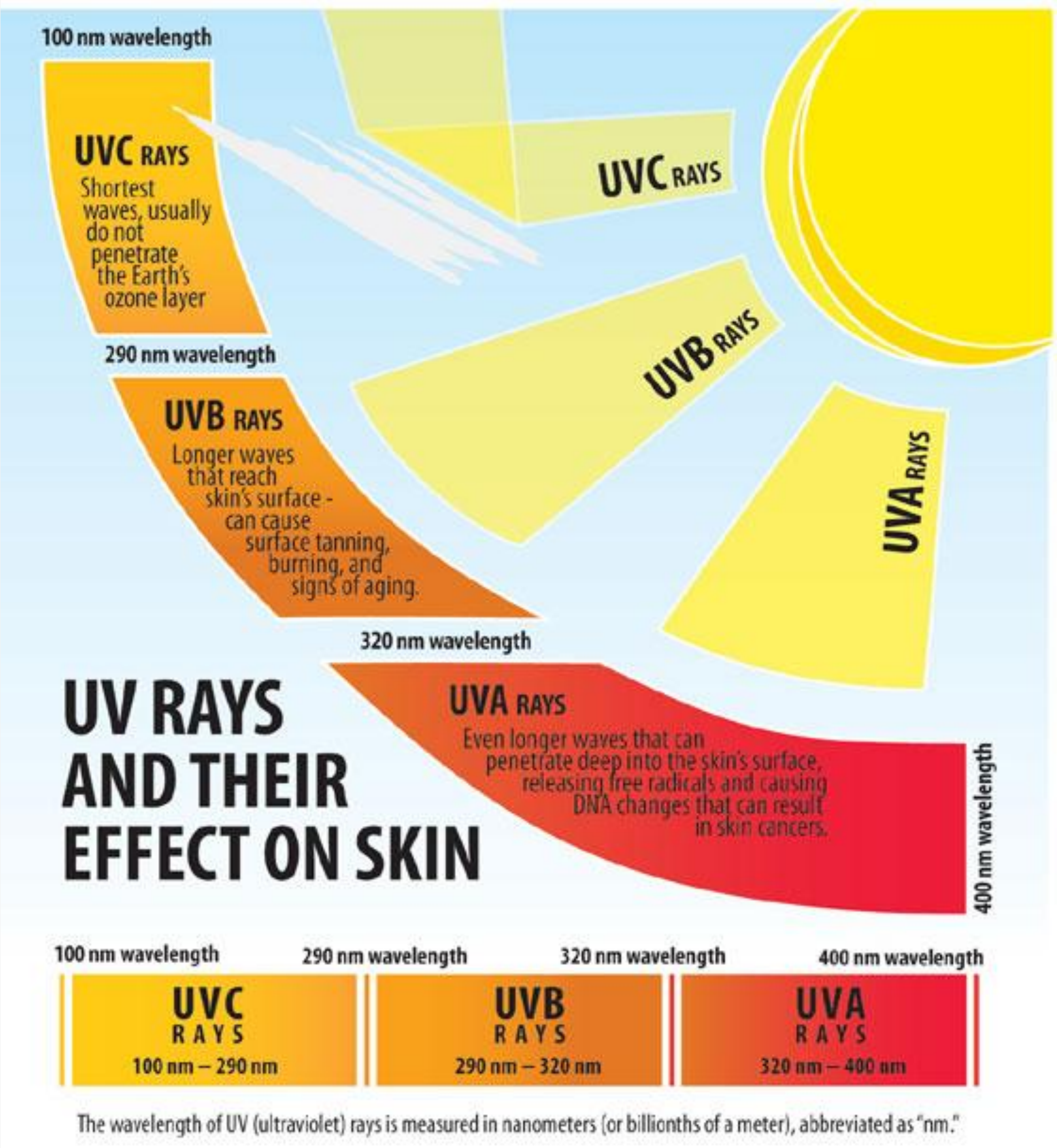
SUNSCREENS

- The American Cancer Society recommends the use of sunscreen because it aids in the prevention of **squamous cell carcinomas**.

- However, many sunscreens **only block UVB radiation**, and therefore do **not block UVA radiation, which is involved in photoaging and cancers**.

- Specifically, UVA does not primarily cause sunburn but rather can increase the rate of **melanoma** and **photodermatitis.**

- The use of broad-spectrum (UVA/UVB) sunscreens can address this concern.



PHOTOBLOCKING INORGANIC FILTERING AGENTS

- photoblocking/inorganic filtering (“Mineral Sunscreens”)
  - aka “physical” or “mineral” photoprotectants
  - Zinc Oxide
    - opaque white
  - Titanium Dioxide
    - opaque white
  - Wide range of wavelengths blocked (“broad-spectrum”).
    - particularly ZnO_2
    - note also iron oxides
      - various yellow, brown, red, ochre, black pigments

Zn^{+2} O^{-2}

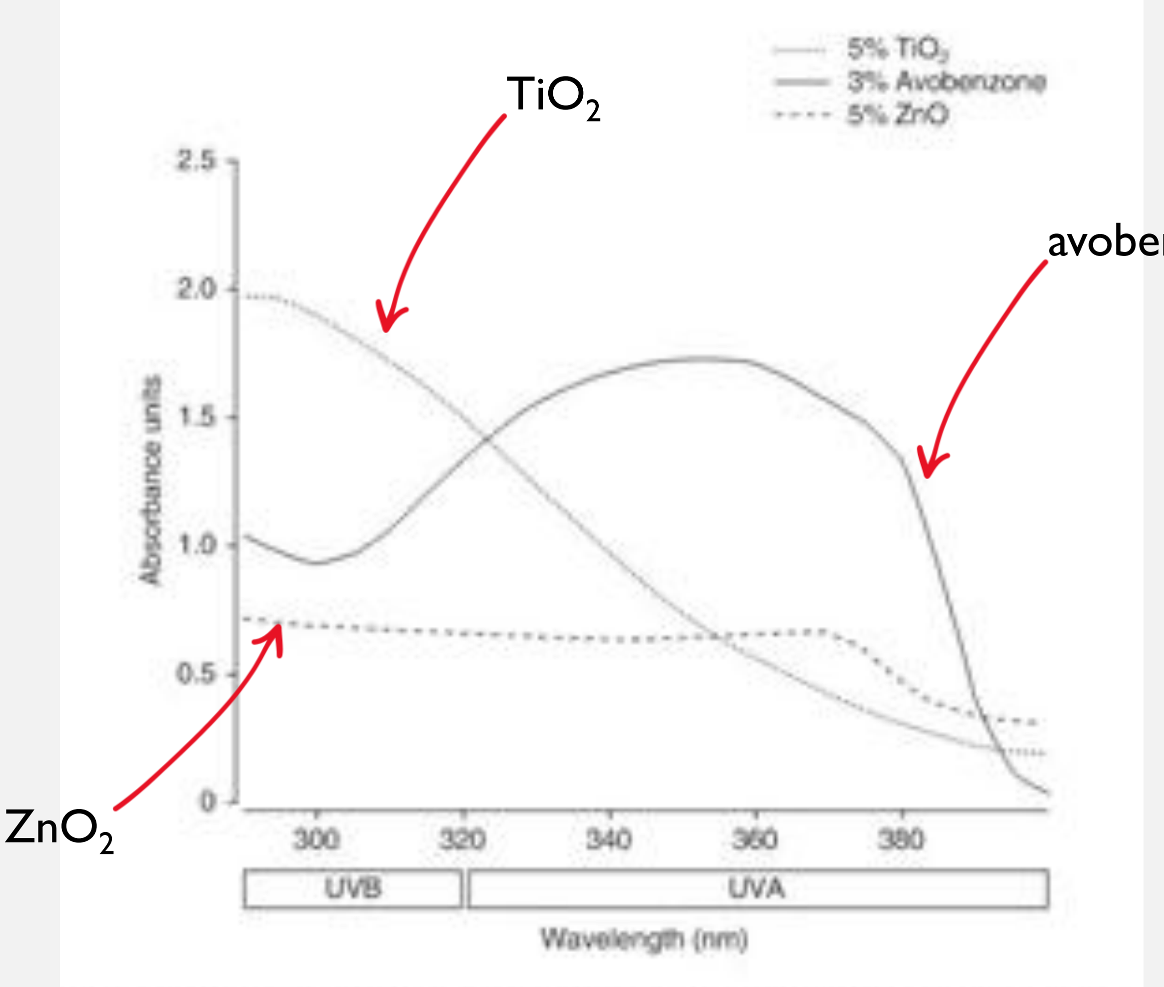
zinc oxide

O^{2-} Ti^{4+} O^{2-}

titanium dioxide

labmuffin.com

INORGANIC FILTERING



control/standardizing wavelength

Fig. 1. Comparison of UV attenuation profiles for avobenzone, zinc oxide (ZnO), and titanium dioxide (TiO_2) as single ingredients in a model formulation.

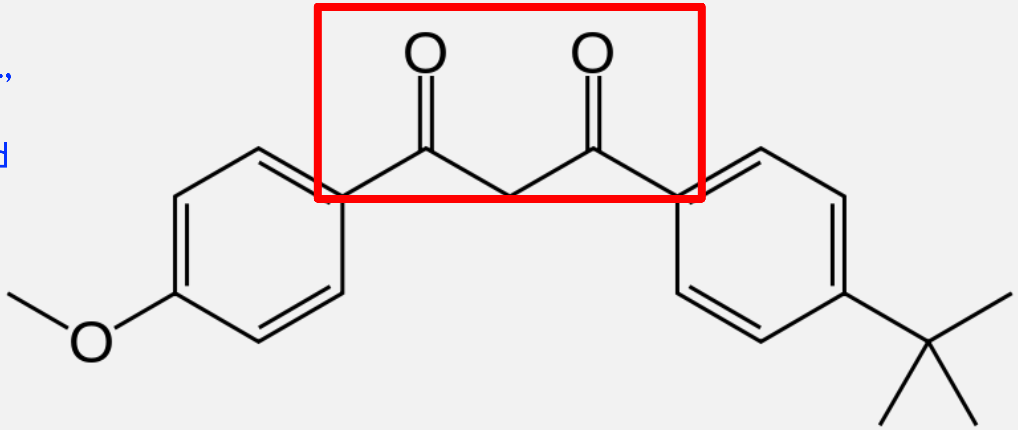
ORGANIC FILTERING AGENTS

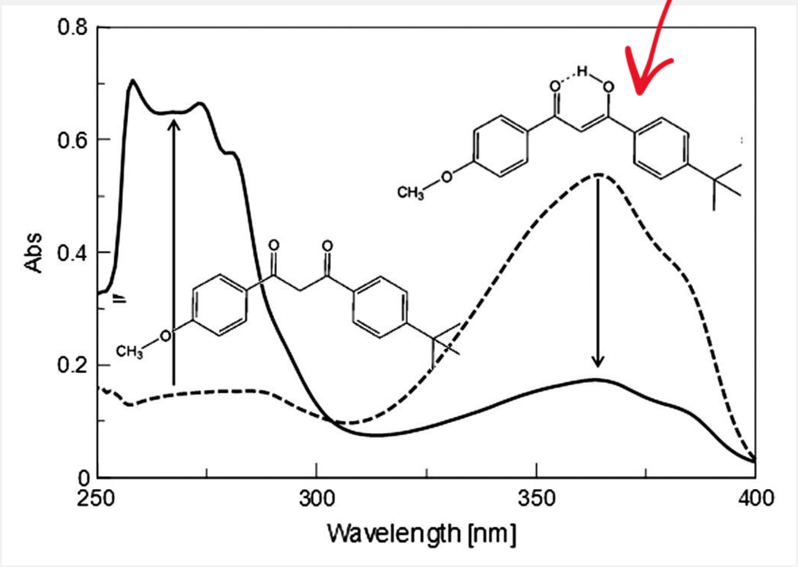
- a.k.a., chemical sunscreens
  - **Avobenzone** (1-(4-Methoxyphenyl)-3-(4-*tert*-butylphenyl)propane-1,3-dione) exists in the ground state as a mixture of the enol and keto forms, favoring the chelated enol. This enol form is stabilized by **intramolecular hydrogen-bonding** within the β-diketone. Absorbs ultraviolet light **over a wider range of wavelengths (UVA and UVB)** than many sunscreen agents. Marketed as "broad spectrum" sunscreens. Has absorption maximum of 357 nm.

What's the point here???

electron resonance - absorb energy and turn into heat

This and next 5 slides: the most common chemical sunscreen agents in the U.S., Note opportunities for resonance that predict good UV absorbance due to ability to redistribute e^-

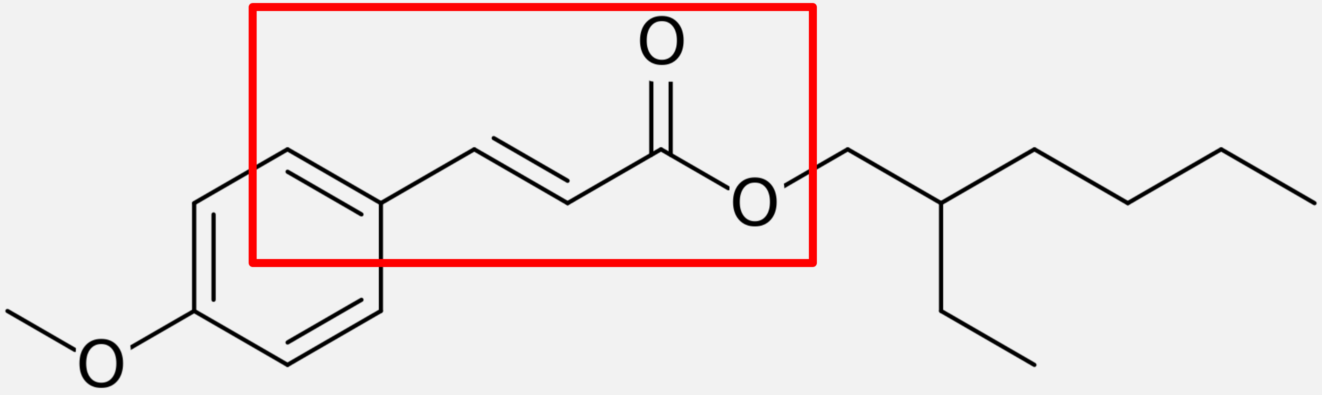


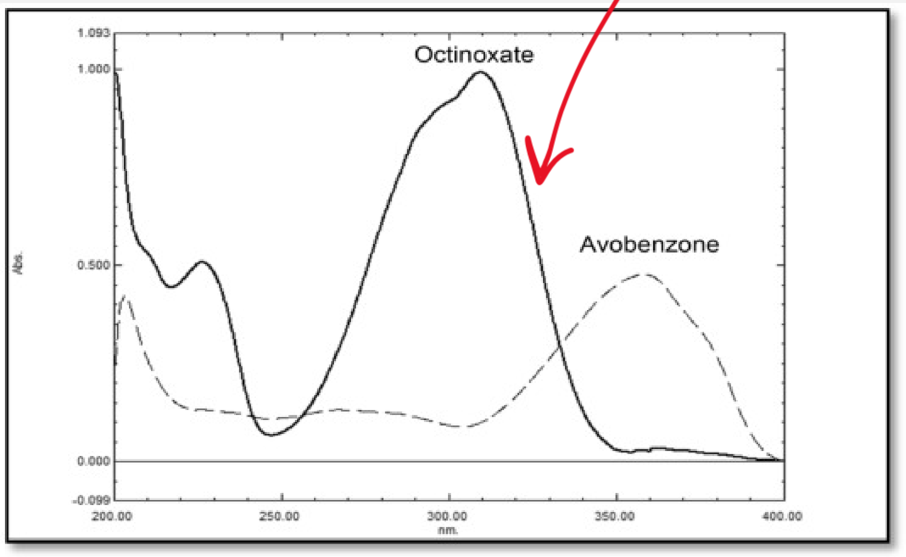


J. Photochem. Photobiol. B, **140**, 36-40

FYI

- **Organic Filtering ("Chemical Sunscreens")**
  - **Octinoxate** (Octyl methoxy**cinnamate** or ethylhexyl methoxy**cinnamate**) is an ester formed from methoxycinnamic acid and (RS)-2-ethylhexanol. Used as an active ingredient in sunscreens combined with **oxybenzone** and **titanium oxide** for its use in protection against UV-B rays.



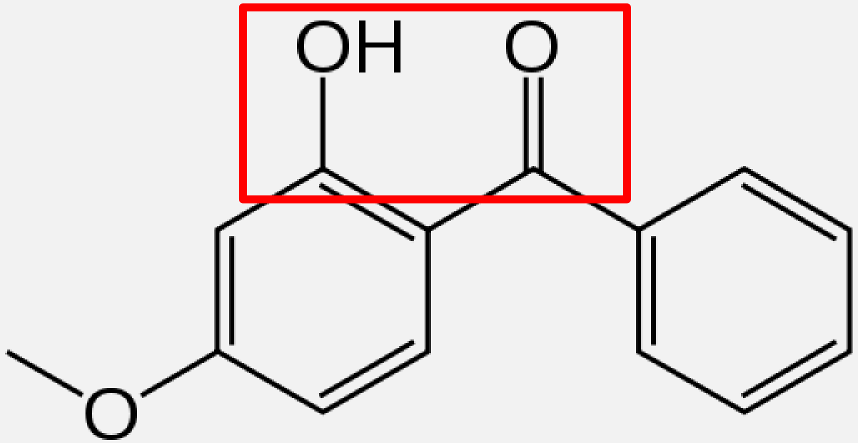


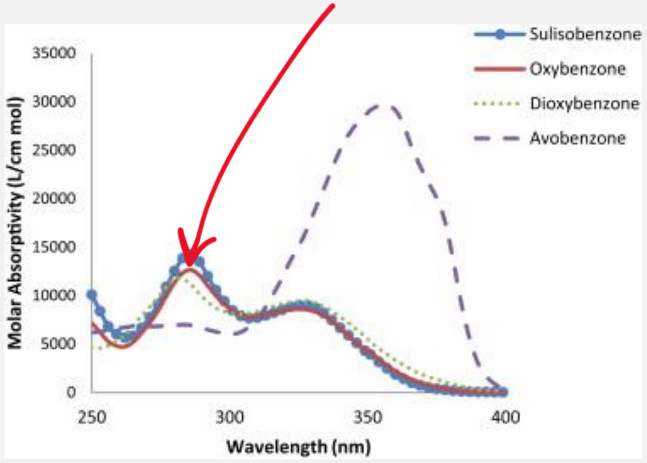
https://doi.org/10.1016/j.saa.2018.07.073

FYI

ORGANIC FILTERING AGENTS CONT'D

- **Organic Filtering ("Chemical Sunscreens")**
  - **Oxybenzone** (2-Hydroxy-4-methoxyphenyl)-phenylmethanone) provides broad-spectrum ultraviolet coverage, **including UVB and short-wave UVA rays**. It has an absorption profile spanning from 270 to 350 nm with absorption peaks at 288 and 350 nm. Because it is a **conjugated** molecule, oxybenzone absorbs at lower energies than many aromatic molecules. Also, the **hydroxyl group is hydrogen bonded to the ketone**, and this contributes to oxybenzone's light-absorption properties. It is one of the most widely used organic UVA filters in sunscreens today.



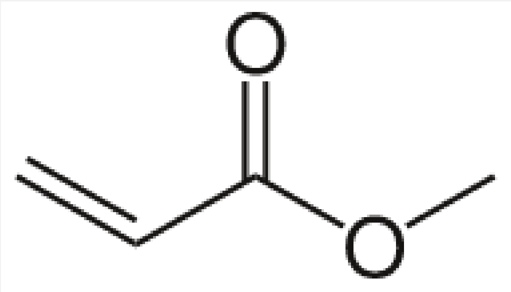


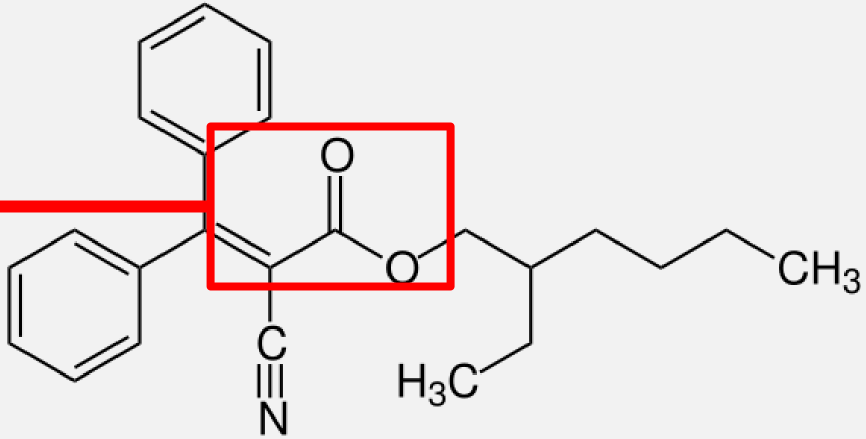
https://doi.org/10.1016/j.ijpharm.2015.04.027

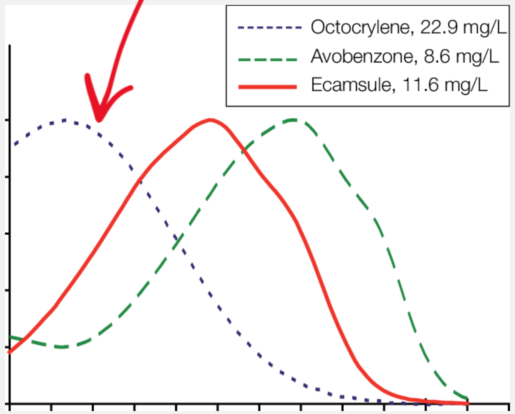
FYI

ORGANIC FILTERING AGENTS CONT’D

- **Organic Filtering (“chemical sunscreens”)**
  - **Octocrylene** is an ester formed by the reaction of 3,3-diphenylcyanoacrylate with 2-ethylhexanol. The **acrylate** portion of the molecule absorbs UVB **and** short-wave UVA (ultraviolet) rays with wavelengths from 280 to 320 nm, protecting the skin from direct DNA damage. Stabilizes avobenzone in combined formulation





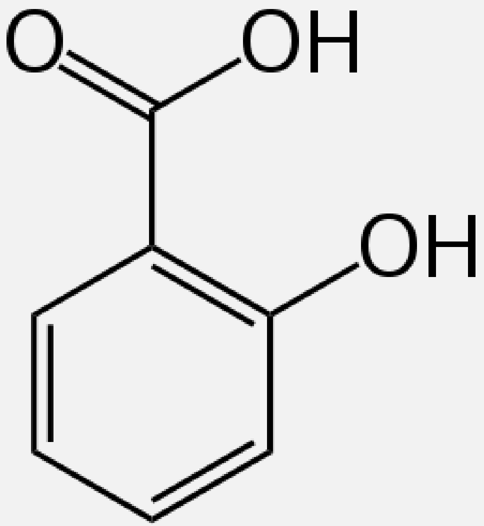


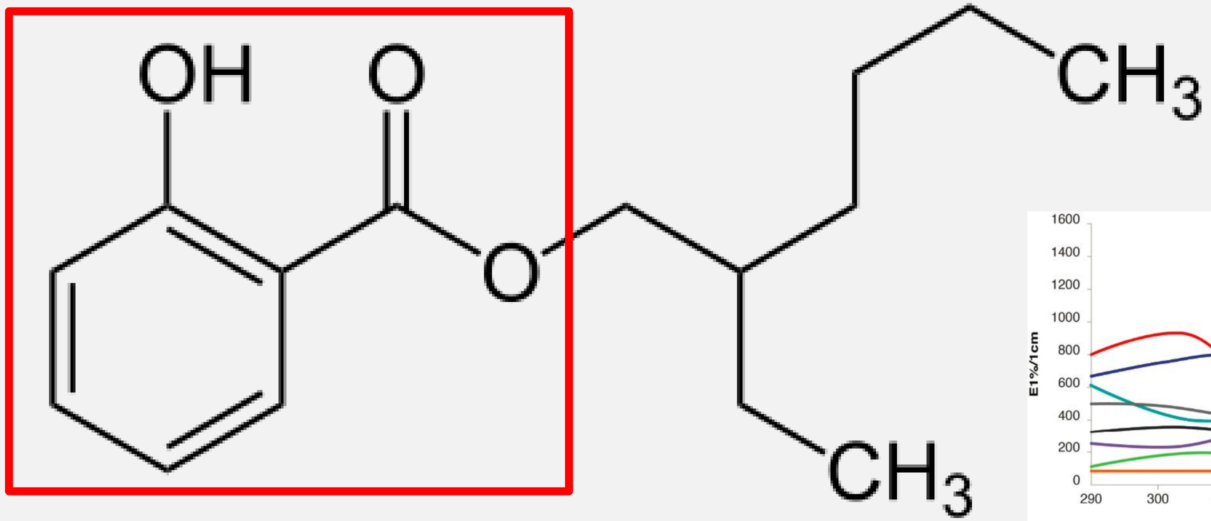
Cutis 83(2):95-103

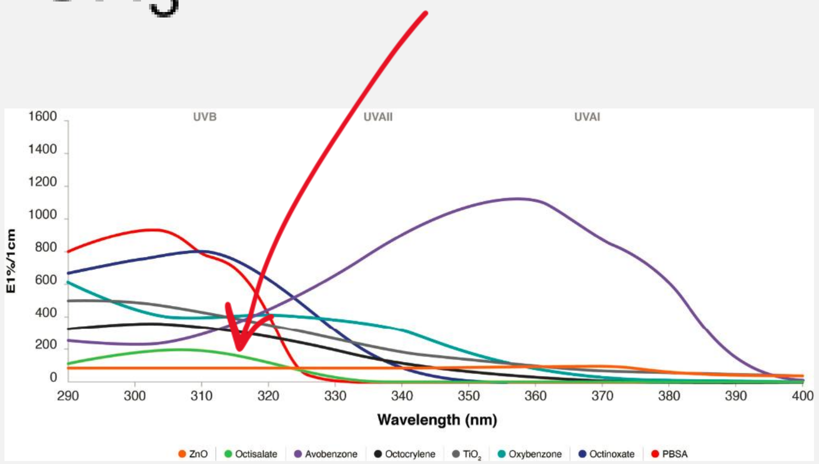
FYI

ORGANIC FILTERING AGENTS CONT'D

- **Organic Filtering ("Chemical Sunscreens")**
  - **Octisalate** (Octyl salicylate, or 2-ethylhexyl salicylate)**,** is an ester formed by the condensation of a salicylic acid with 2-ethylhexanol and is used as an ingredient in sunscreens to absorb UVB (ultraviolet) rays. The **salicylate** portion of the molecule absorbs ultraviolet light with a wavelength from 295 nm to 315 nm.



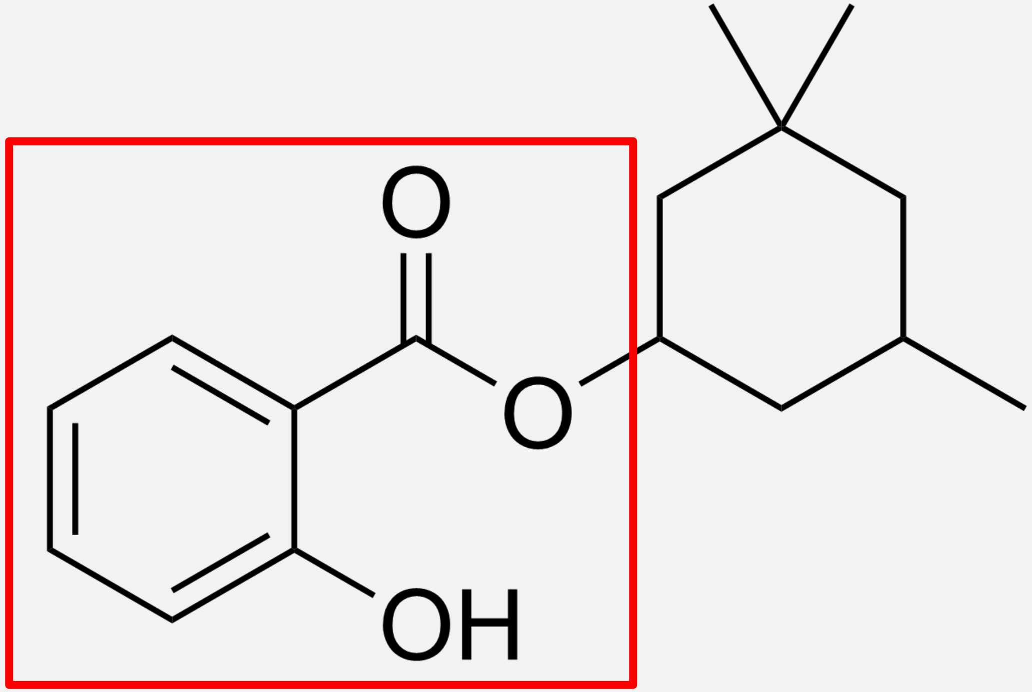


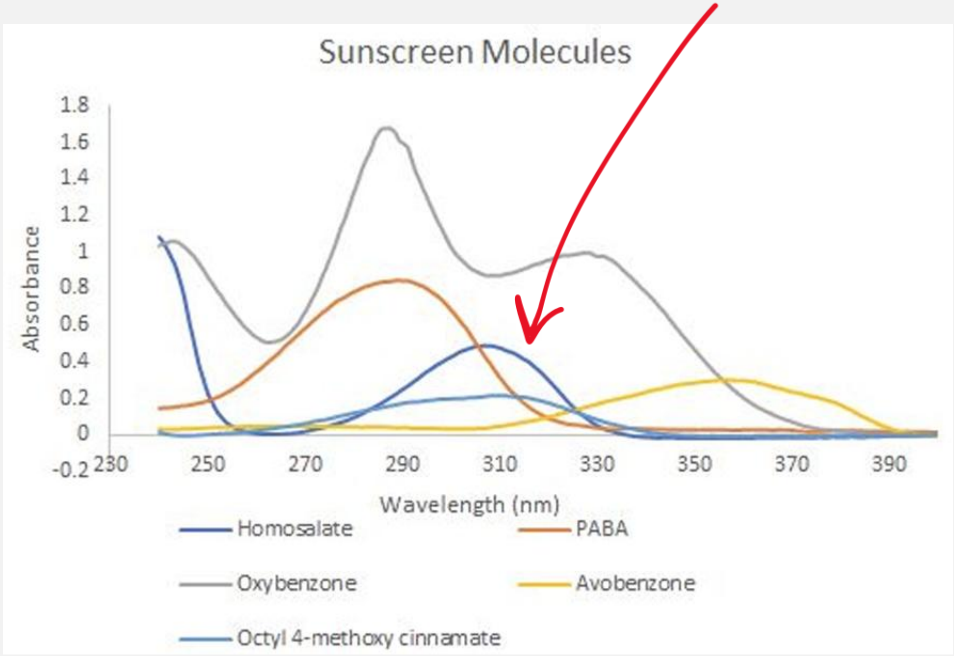


FYI

ORGANIC FILTERING AGENTS CONT'D

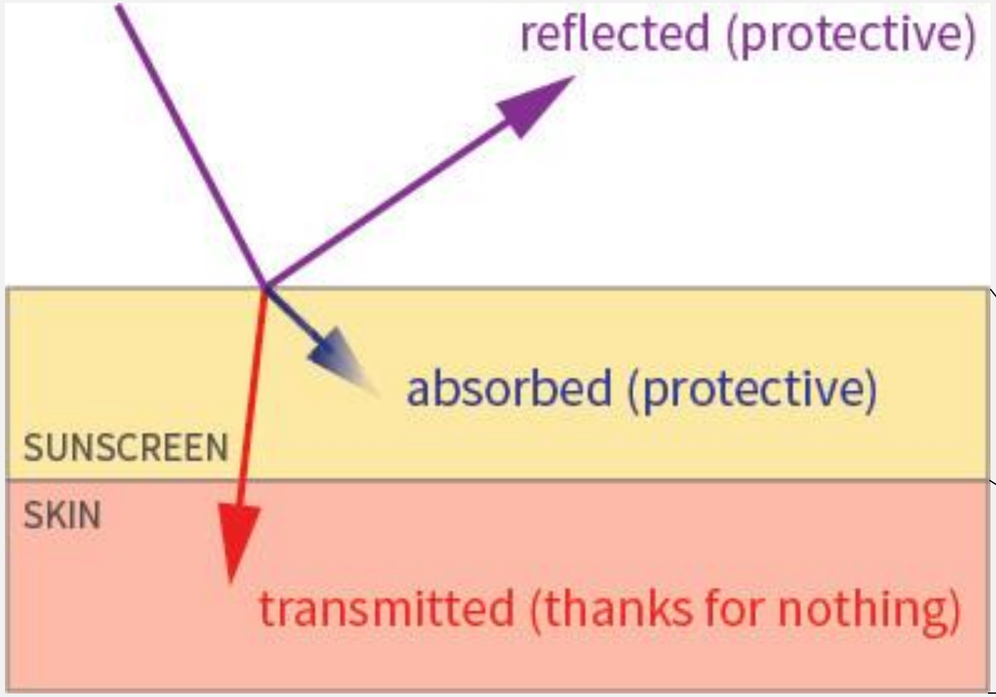
- **Organic Filtering ("Chemical Sunscreens")**
  - **Homosalate** (3,3,5-Trimethylcyclohexyl 2-hydroxybenzoate) is an ester formed from salicylic acid and 3,3,5-trimethylcyclohexanol, a derivative of cyclohexanol. It is **contained in 45% of U.S. sunscreens**. The **salicylic acid** portion of the molecule absorbs ultraviolet rays with a wavelength from 295 nm to 315 nm, protecting the skin from sun damage.

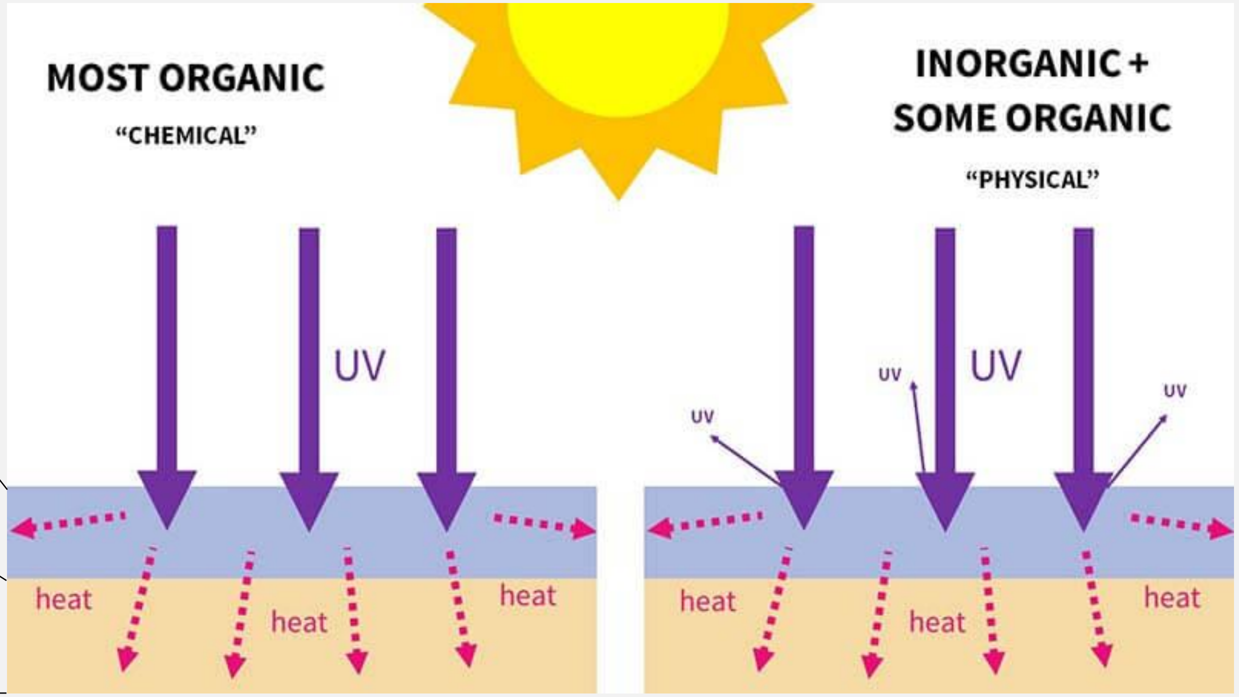




FYI

PHYSICAL AND CHEMICAL SUNSCREENS BOTH ABSORB UV RAYS





TiO_2 and ZnO_2 reflect about 4% of UV exposure

take-home: the MOA of both chemical and physical sunscreens is mostly UV absorption

https://doi.org/10.1111/phpp.12214

BEWARE MISINFORMATION

https://lowchailing.com/2019/05/20/chemicals-in-sunscreen-may-enter-blood/

This is the blog of a physician, who should know better.

Grains of truth: organic filters can cross the skin barrier of frozen/thawed pig's ear, a common model system for human skin product testing. separately, rats fed about 1% of their bodyweight in sunscreen over time displayed responses consistent with estrogen signaling.

Misrepresented/Unaddresed: To what extent does this apply to intact human skin in vivo? What are the known biological effects of such absorption in humans applying suncreen to their skin, not eating 1-2 pounds per day? How does organic filter absorption into the skin compare to mineral absorbtion?

Take-home: be skeptical of sweeping claims and categorizations. Note lack of context and lack of scale. Note sensationalizing language ("chemicals").

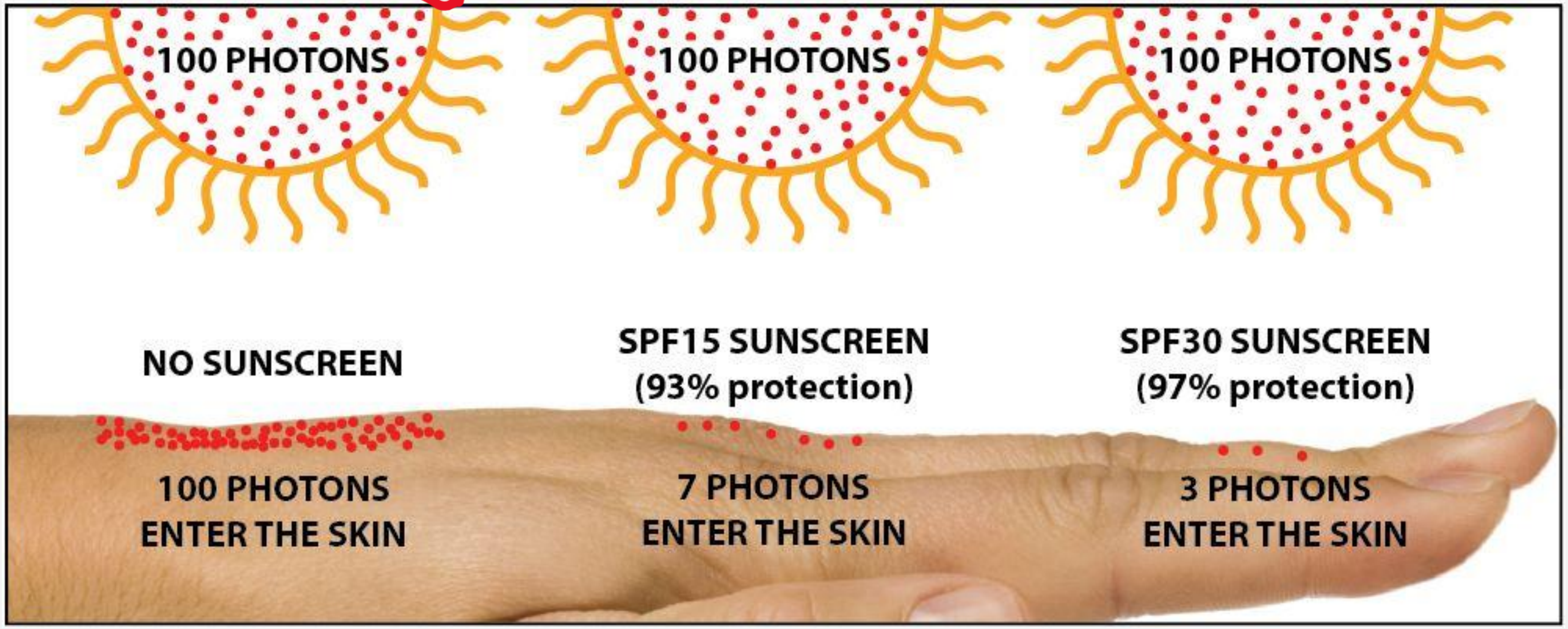
Side note: the same paper cited above states: "Escalation of a phobia towards all organic UV filters is undesirable". The exact opposite of what is happening in this image.

FYI

SUN PROTECTION FACTOR SIMPLIFIED

SPF-15-30-50 Guide:

- **15**- 1/15 of the UVB rays get through to your skin - blocking about 93%.
- **30**- 1/30 of the UVB rays get through to your skin - blocking about 97%.
- **50**- 1/50 of the UVB rays get through to your skin - blocking about 98%.

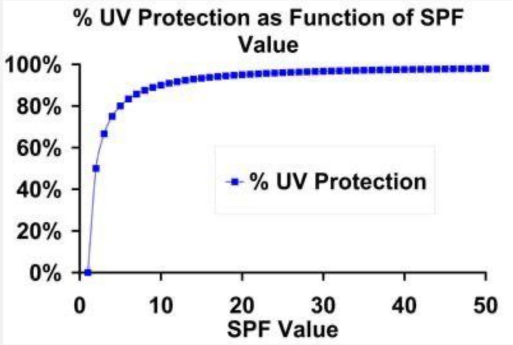


SPF **15** = 93%

SPF **30** = 97%

SPF **50** = 98%

SPF **70** = 98.5%



diminishing returns
SPF only for UVB

ACTUAL SPF CALCULATION: IT'S MORE COMPLICATED

- SPF ("Sun Protection Factor") is a **complex** relationship: oversimplification can lead to inaccuracy.
  - Specifically, the EAS (erythemal action spectrum, a measure of tissue damage)
  - also note that testing conditions do not match typical use: much MORE product used in testing than people tend to apply in real life
    - 2-2.5 mL recommended for face and neck; typical use is half or less recommended amt.

- The most important thing to know is that **SPF15 is fine** and SPF 30 will only give you about 4% more protection.

- One the other hand, remembering to reapply about every 2 hours, and picking broad spectrum formulations are important.

know

calc is FYI

Mathematically, the SPF (or the UPF) is calculated from measured data as

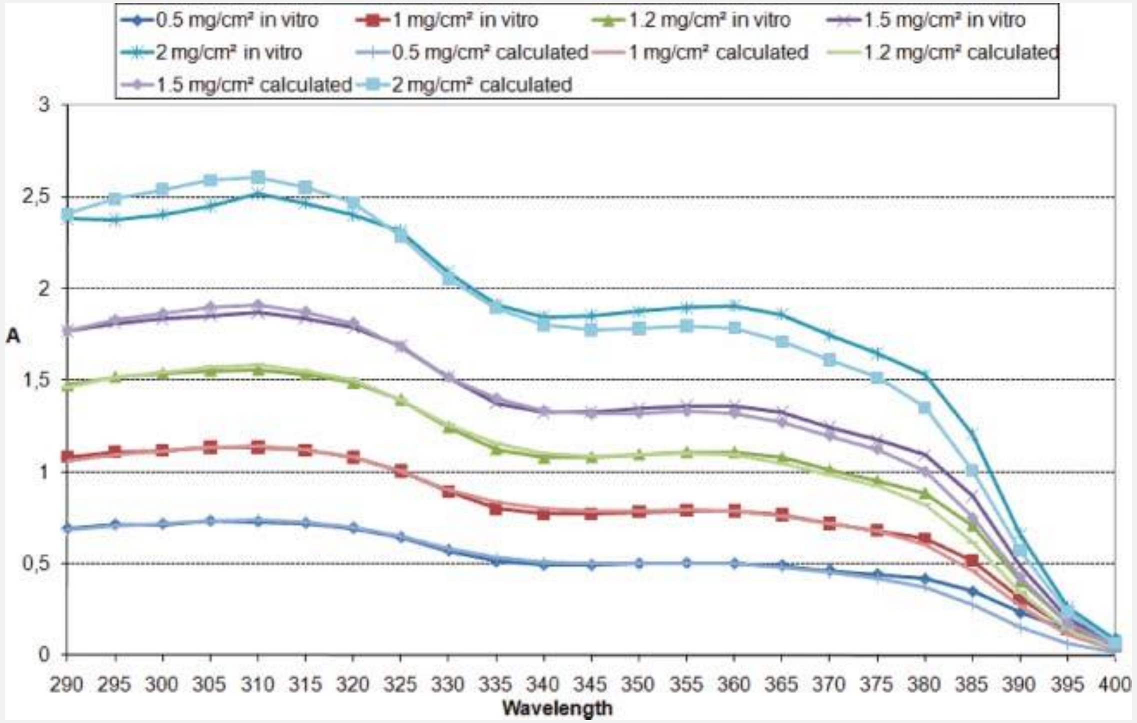
$$\mathrm{SPF} = \frac{\int A(\lambda)E(\lambda)d\lambda}{\int A(\lambda)E(\lambda)/\mathrm{MPF}(\lambda)\,d\lambda},$$

where $E(\lambda)$ is the solar irradiance spectrum, $A(\lambda)$ the erythemal action spectrum, and $\mathrm{MPF}(\lambda)$ the monochromatic protection factor, all functions of the wavelength λ. The MPF is roughly the inverse of the transmittance at a given wavelength.

WHY PROPER AND REAPPLICATION IS RECOMMENDED

reapply every 2 hours

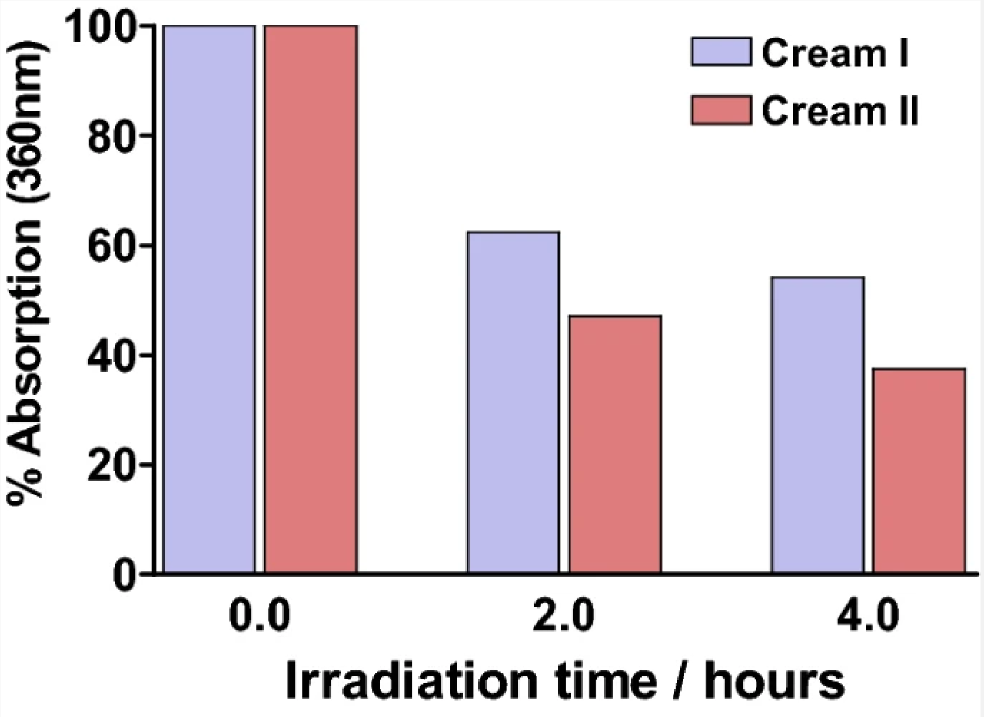
proper application ensures more UV absorbance by the sunscreen



DOI:10.1039/b9pp00183b

UV absorbance of one sunscreen applied at different average thicknesses

UV absorbance by sunscreens diminishes over time



https://link.springer.com/article/10.1007/s43630-021-00013-1

UV absorbance of 2 sunscreens over 4 h (% of 0 h). Cream I: titanium dioxide, ethylhexyl triazone, avobenzone, and octinoxate; Cream II: octyl salicylate, oxybenzone, avobenzone, and octinoxate.

TINTING AND UVA PHOTOPROTECTION WITH IRON OXIDES

106 JOURNAL OF THE SOCIETY OF COSMETIC CHEMISTS

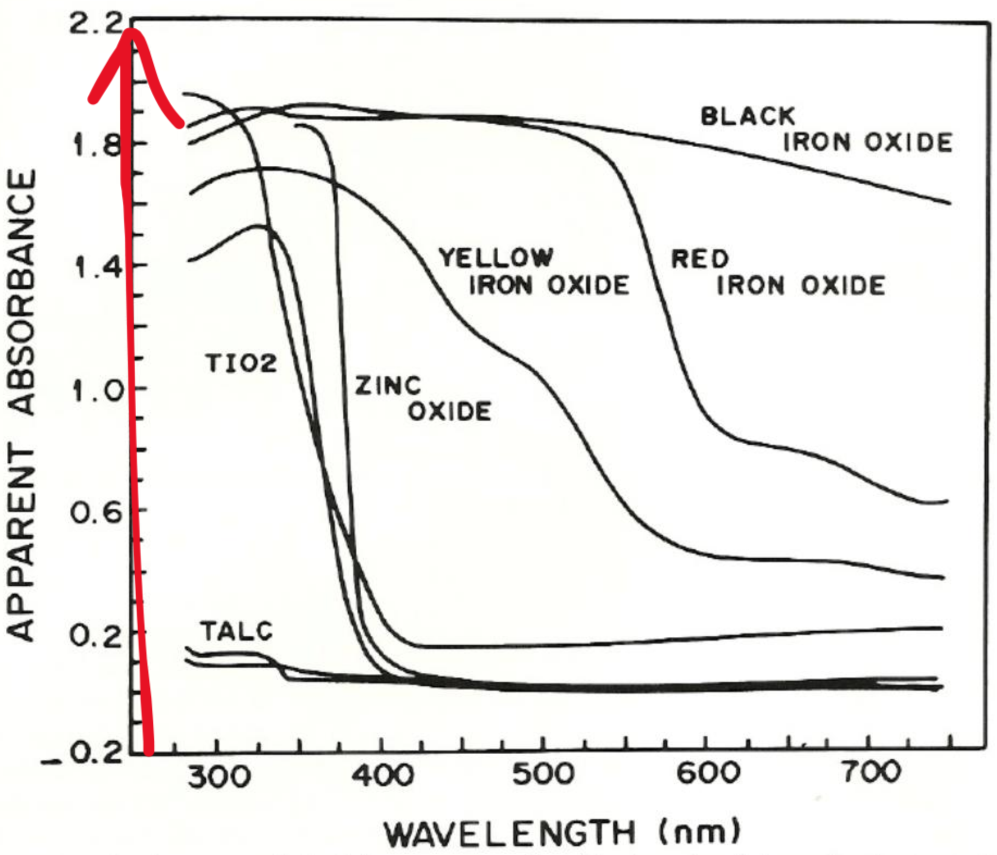
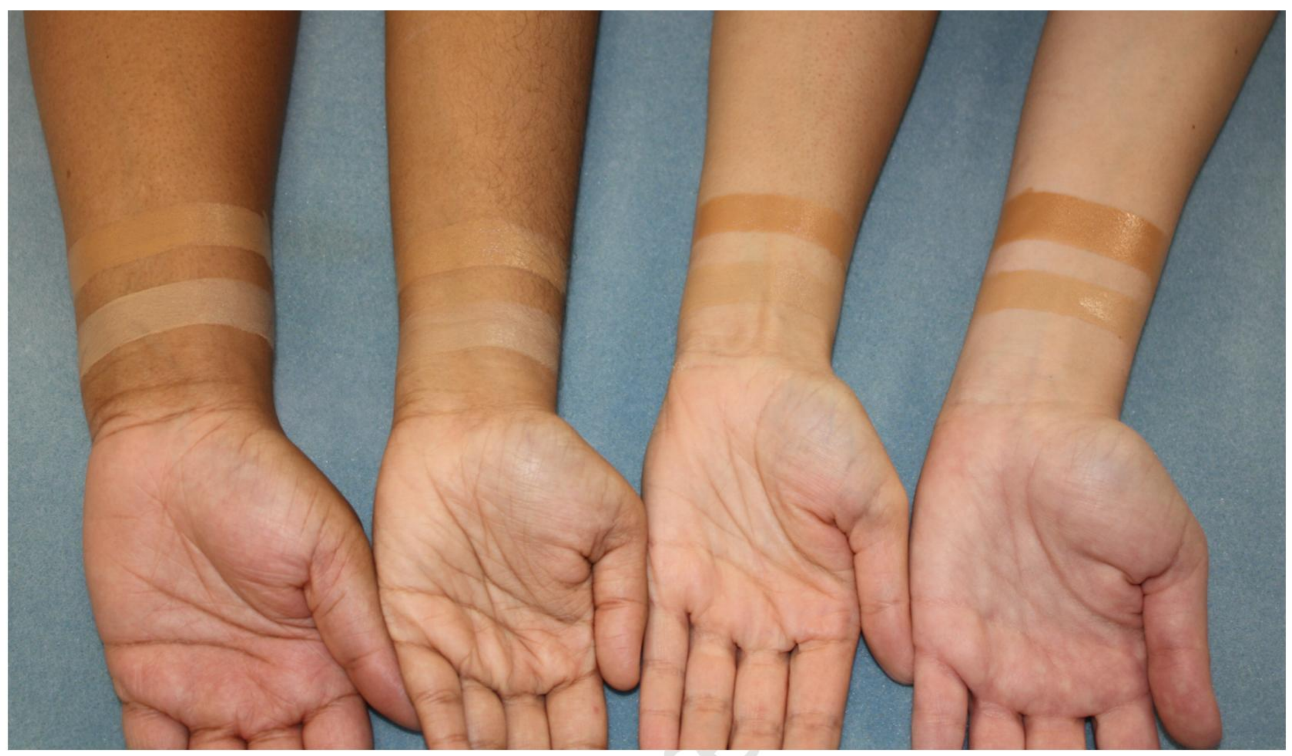


Figure 1. Reflection spectra of physical sunscreen powders. The data plotted shows absorbance vs wavelength. An absorbance of 0.0 corresponds to 100% reflection of the light relative to $BaSO_4$; an absorbance of 1.0 corresponds to 10%; 2.0 corresponds to 1% reflection.

1990 Sayre, et al Journal of the Society of Cosmetic Chemists 41(2):103-109



doi:10.1016/j.jaad.2020.04.079

iron oxides add increased photoprotection in the UVA/vis range AND coloration to sunscreen product

clinical note: rec. for melasma

QUICK REVIEW

- What is the product of a Type I psoralen reaction?
- What are the "ABC"s of UV categories, and how much does the atmosphere protect you from each one?

  A = aging, B - burning, C = captured

- What active ingredients are probably in a broad-spectrum "chemical" sunscreen?

  avobenzome ,homosalate, octocrylene

- What active ingredients are probably in a "tinted mineral" sunscreen?

  iron oxide, zinc oxide, titanium dioxide

- How much UVA protection does SPF 80 confer?

  idk

QUICK REVIEW

- What is the product of a Type I psoralen reaction?
  - cross-linked DNA adducts
- What are the ABCs of UV wavelength categories, and how much does the atmosphere protect you from each one?
  - Aging (very little), Burning (most, about 90%), Captured (virtually all)
- What active ingredients are probably in a broad-spectrum ”chemical” sunscreen?
  - examples: avobenzone, homosalate, octocrylate, etc.
- What active ingredients are probably in a “tinted mineral” sunscreen?
  - TiO_2, ZnO_2, iron oxides
- How much UVA protection does SPF 80 confer?
  - haha, who knows?! SPF only applies to *erythemal* response, which is caused by UVB exposure. SPF provides no information on UVA damage.

THANKS!

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- please reach out by email with questions or to schedule individual/group office time
  - response will be faster during normal business hours